

EP/PP

Cognitive neuroscience

Psychology, Philosophy and Linguistics

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JRF in Applied AI | Lady Margaret Hall

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Tutorial Essay

Discuss our understanding of the brain structures and mechanisms that underlie episodic memories?

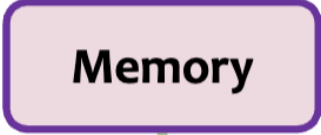
Essay

How have studies of memory disorders helped us to understand memory systems in humans?

- S2024 - What have we learned about episodic memory systems from the study of patients with brain damage?
- M2024 - To what extent can human memory be characterized in terms of the storage and retrieval of information?
- S2023 - What have we learned about episodic memory systems from the study of patients with brain damage?
- S2022 - To what extent can human memory be characterized in terms of the storage and retrieval of information?
- M2021 - What have we learned about episodic memory systems from the study of patients with brain damage?
- M2019 – Human memory is a reconstructive rather than reproductive process. Discuss.
- S2018 – ‘Human memory acts a reconstructive rather than reproductive process.’ Discuss.
- M2018 – How are false memories established? What are the clinical and societal implications?
- S2017 - Why do false memories occur and what impact do they have in legal and other practical contexts?
- M2016 - ‘Human memory acts as a reconstructive rather than reproductive process’. Discuss.
- S2015 - Human memory is a reconstructive rather than reproductive process. Discuss.
- M2015 - Discuss evidence that distinct psychological components are linked to different sections of the serial position curve in human memory research.
- S2014 - Why do false memories occur? What are the practical implications?
- M2013 - What do ‘memory illusions’ tell us about the nature of human memory and what are the practical implications?
- S2012 - To what extent can one characterise human memory in terms of the storage and retrieval of information? What are false memories?
- S2011 - What evidence suggests that people exhibit "false memories"? Why do these false memories arise?
- M2010 - What three psychological processes form the basis of human memory? In your answer give at least one example of an experiment or test where any of these three processes are measured.
- J2009 - What are encoding, storage and retrieval in memory? Give one example of an experiment or test in which one of these processes is measured.
- M2009 - Describe ‘working memory’ and an experiment to measure it.

Different subsystems of memory

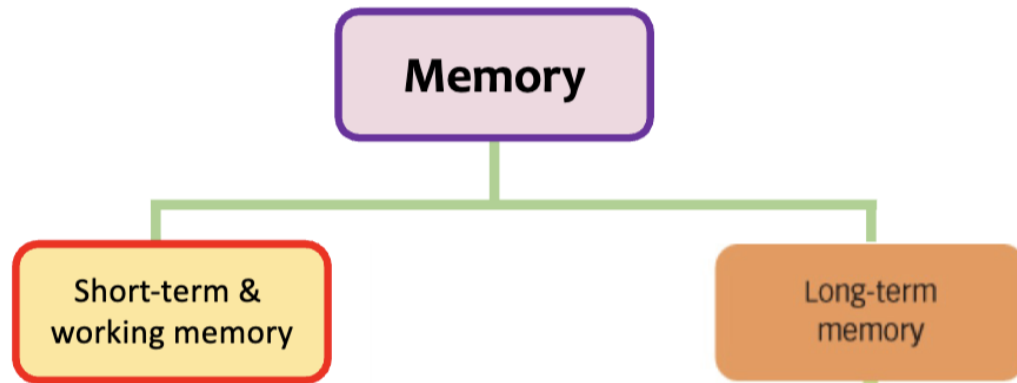
Fractionation of memory systems



Memory

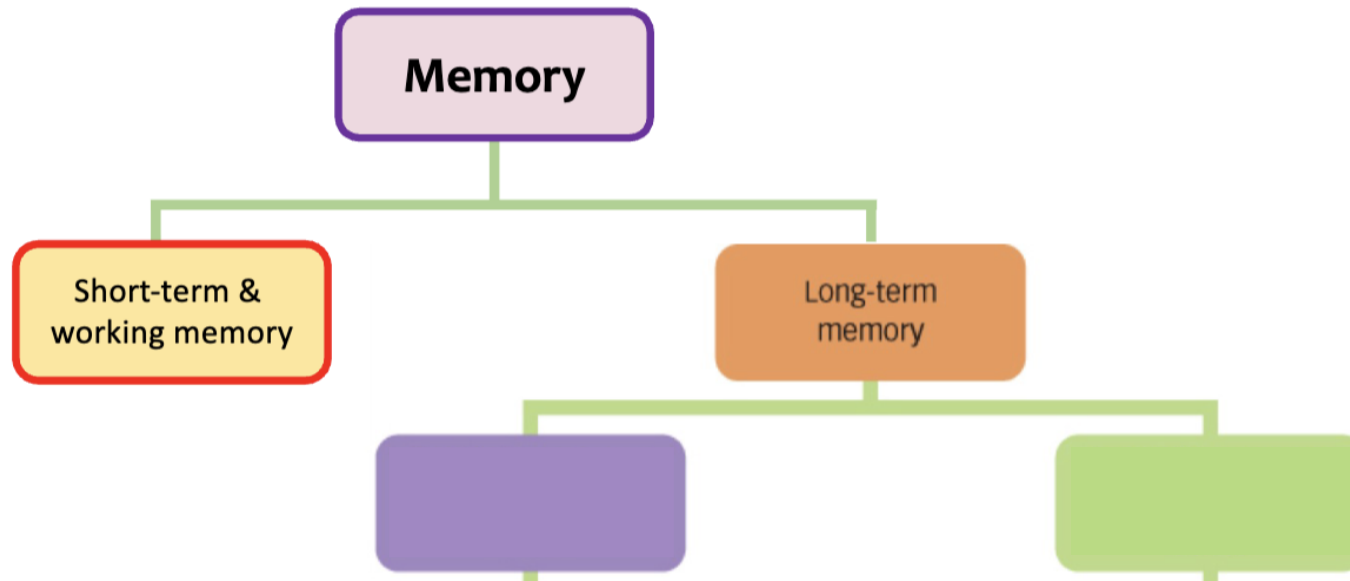
Different subsystems of memory

Fractionation of memory systems



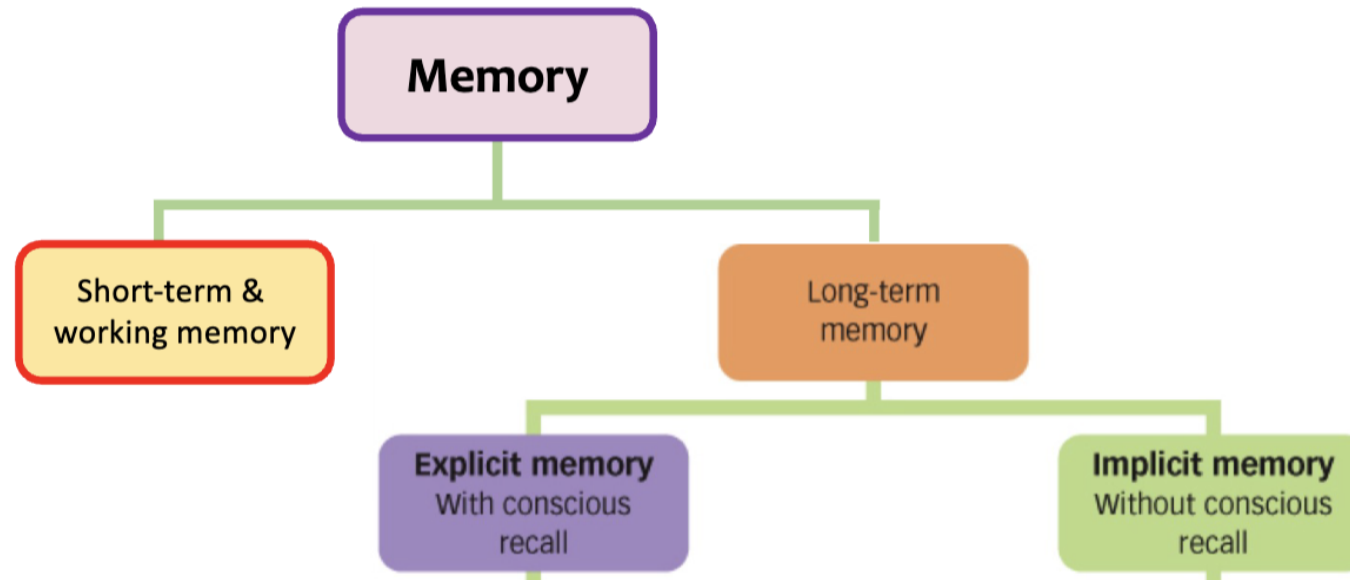
Different subsystems of memory

Fractionation of memory systems



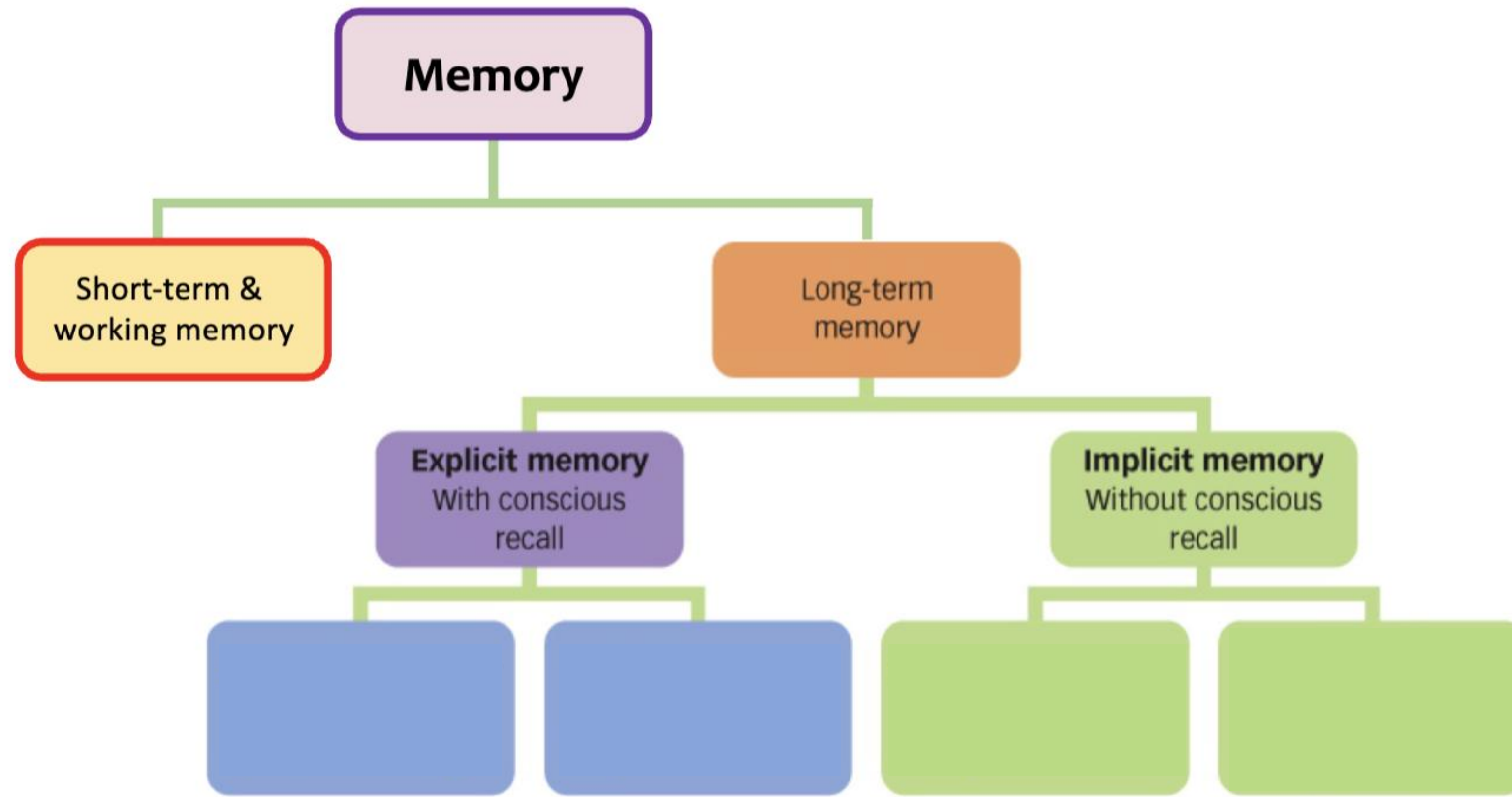
Different subsystems of memory

Fractionation of memory systems



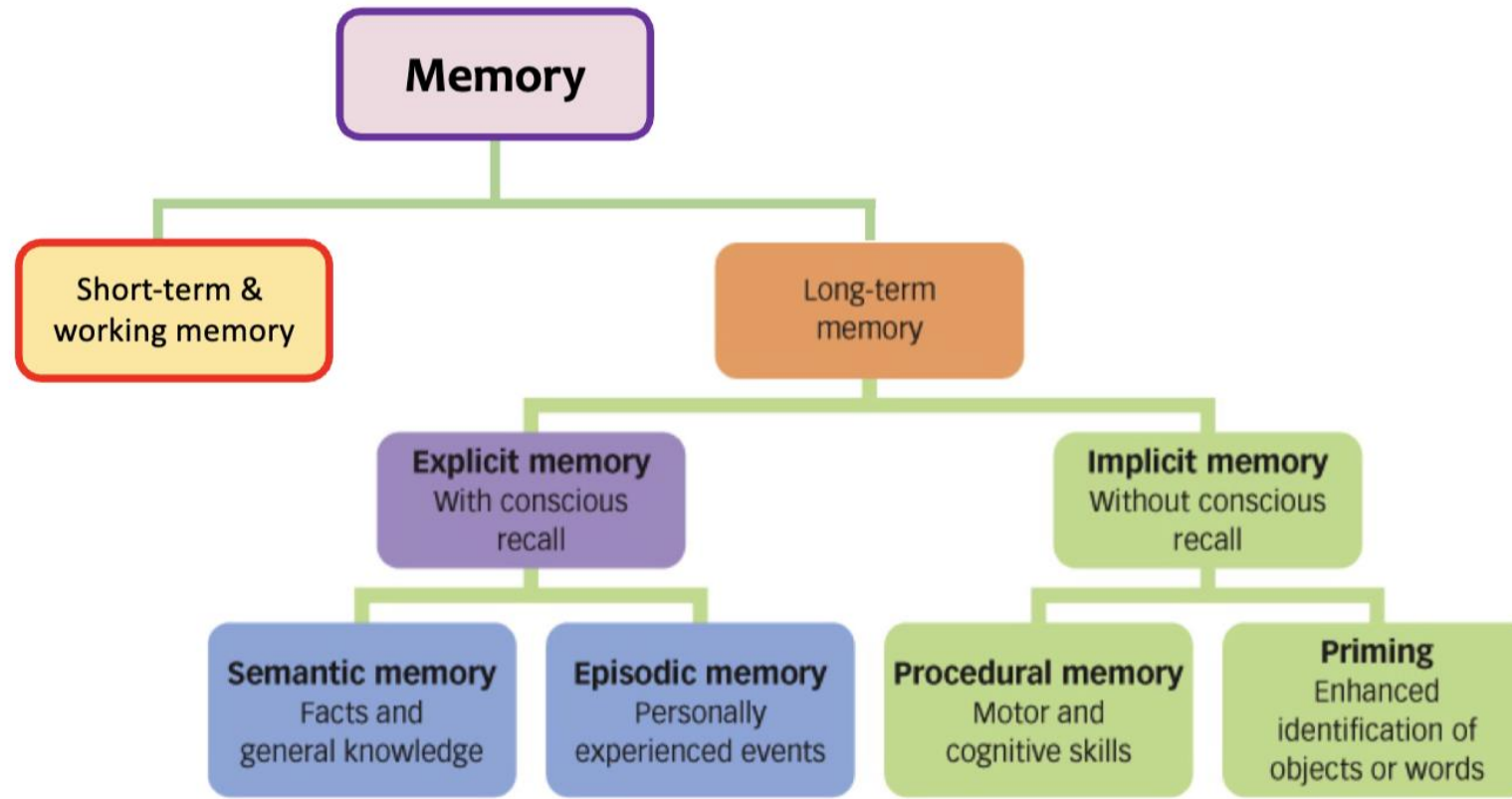
Different subsystems of memory

Fractionation of memory systems



Different subsystems of memory

Fractionation of memory systems



Essay

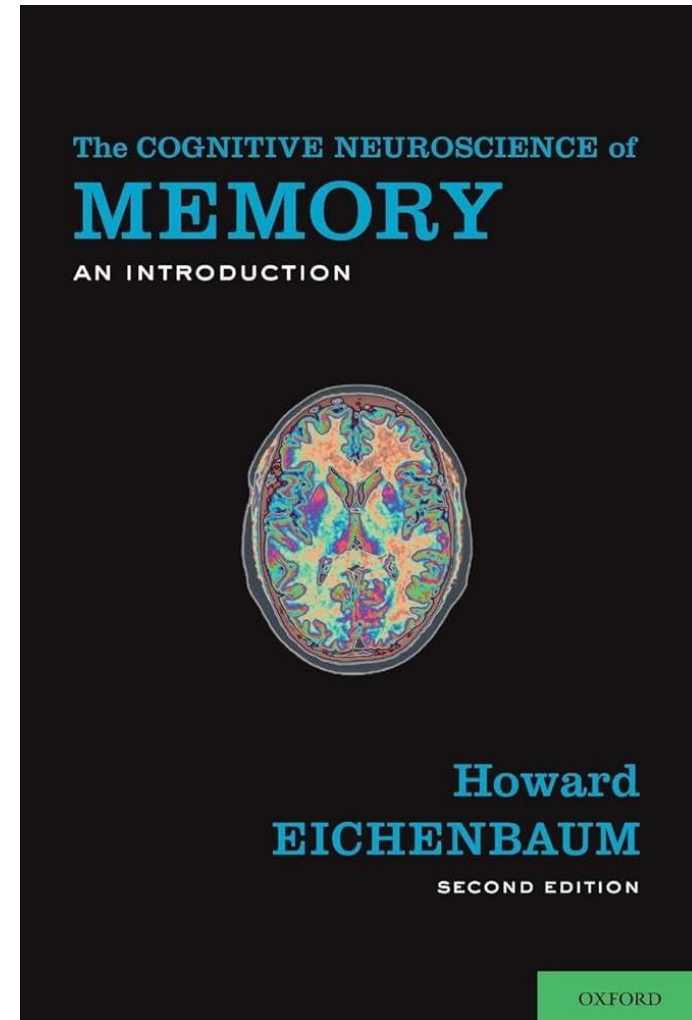
- Definition of memory
- Cognitive tests of memory
- Animal models of memory
- Clinical Cases – (H.M.; learning from amnesia)
- Approach to memory
- Cellular level of memory
- Short-term memory
- Long term memory

Cognitive neuroscience

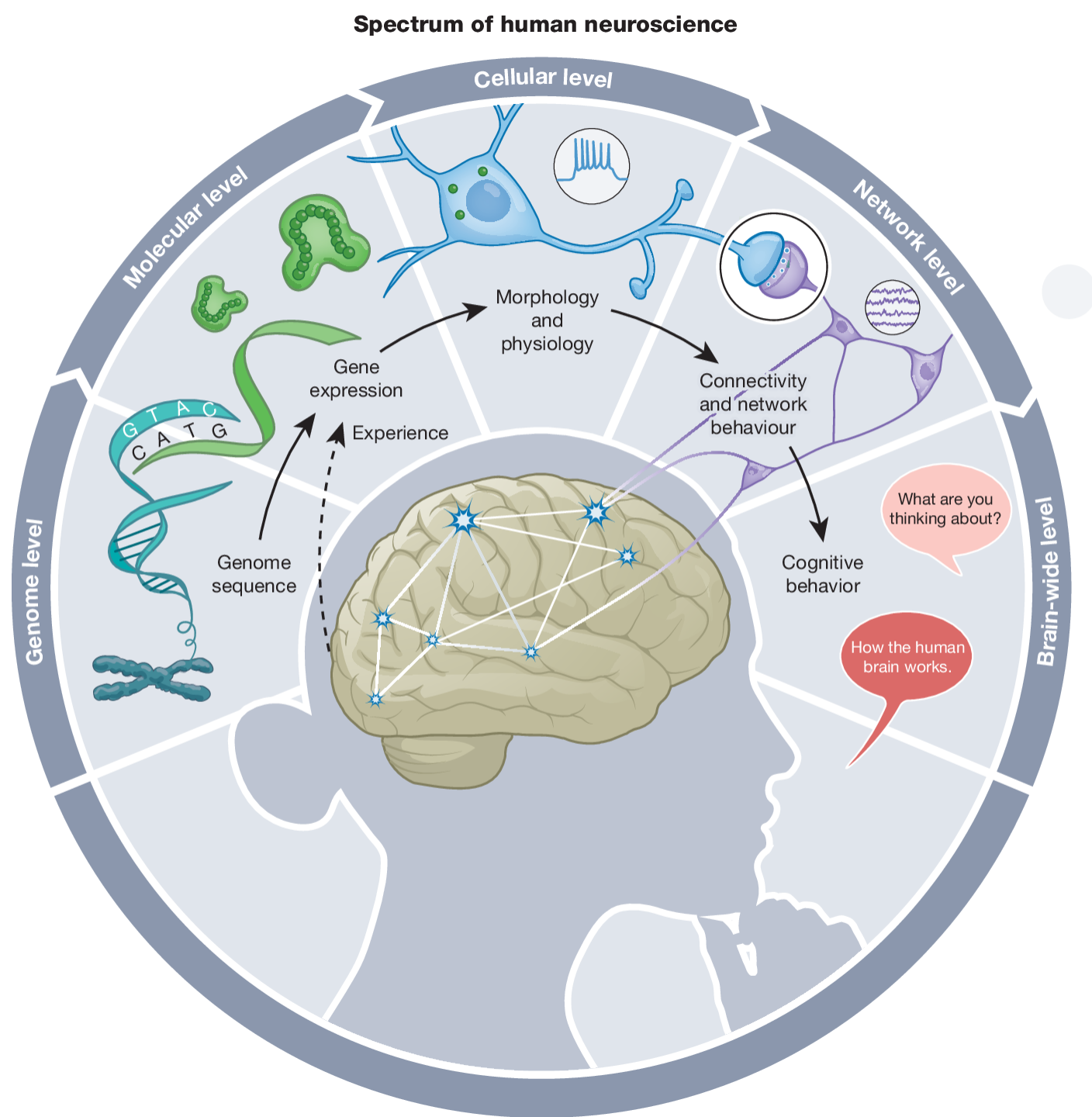
4 Cs of Memory

4 C's of memory:

1. **Connection** – cellular level of memory
2. **Cognition** – memories at a psychological level. Includes behaviorism (all learning is 2/2 conditioned responses) v. cognitivism (complex phenomena such as insight and inference required for complex learning)
3. **Compartmentalisation** – memory is distributed in wide but discrete areas of the brain
4. **Consolidation** – are memories at first labile, then become resistant to loss?

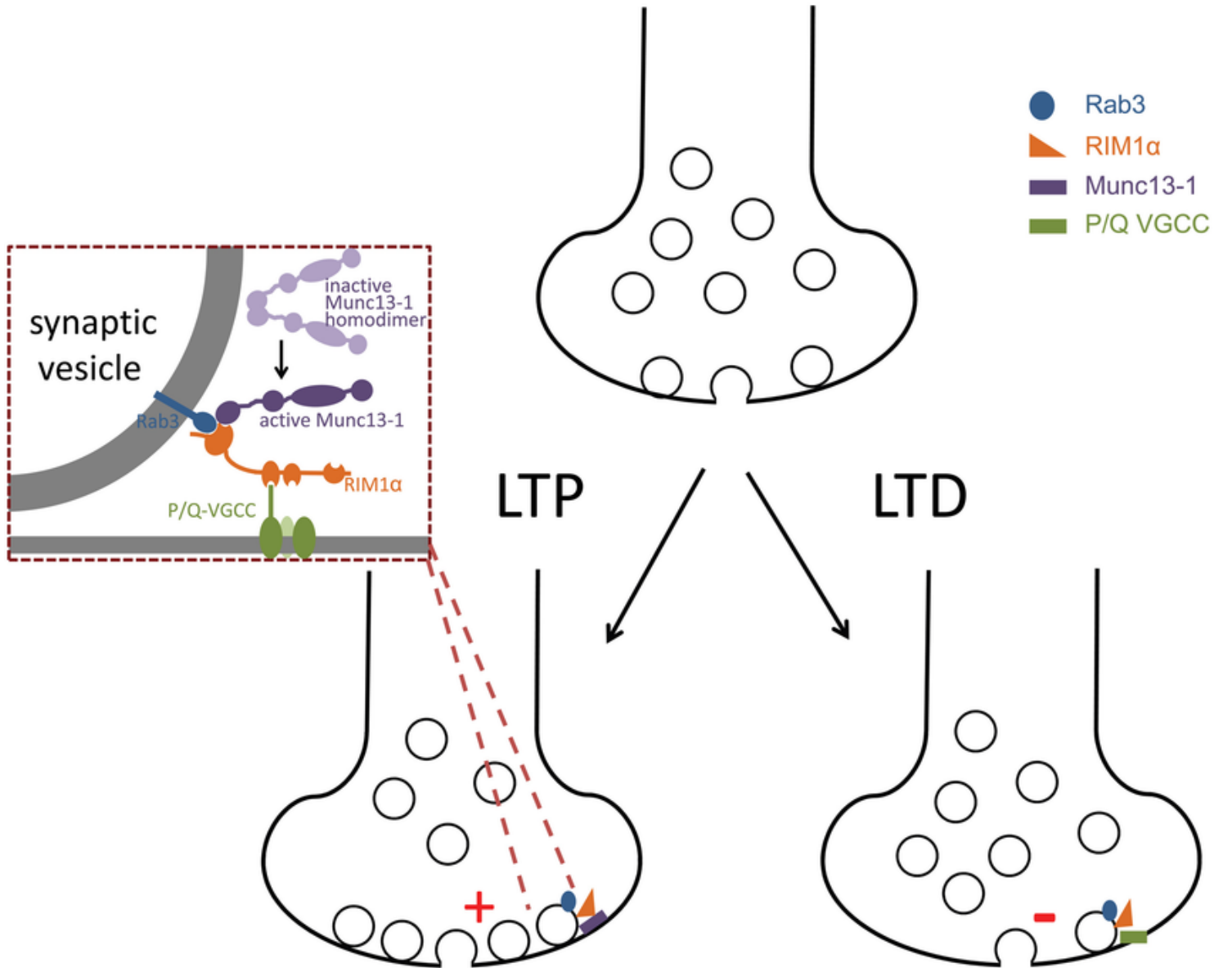


Memory across scales



Cellular memory

synaptic plasticity



Cellular memory

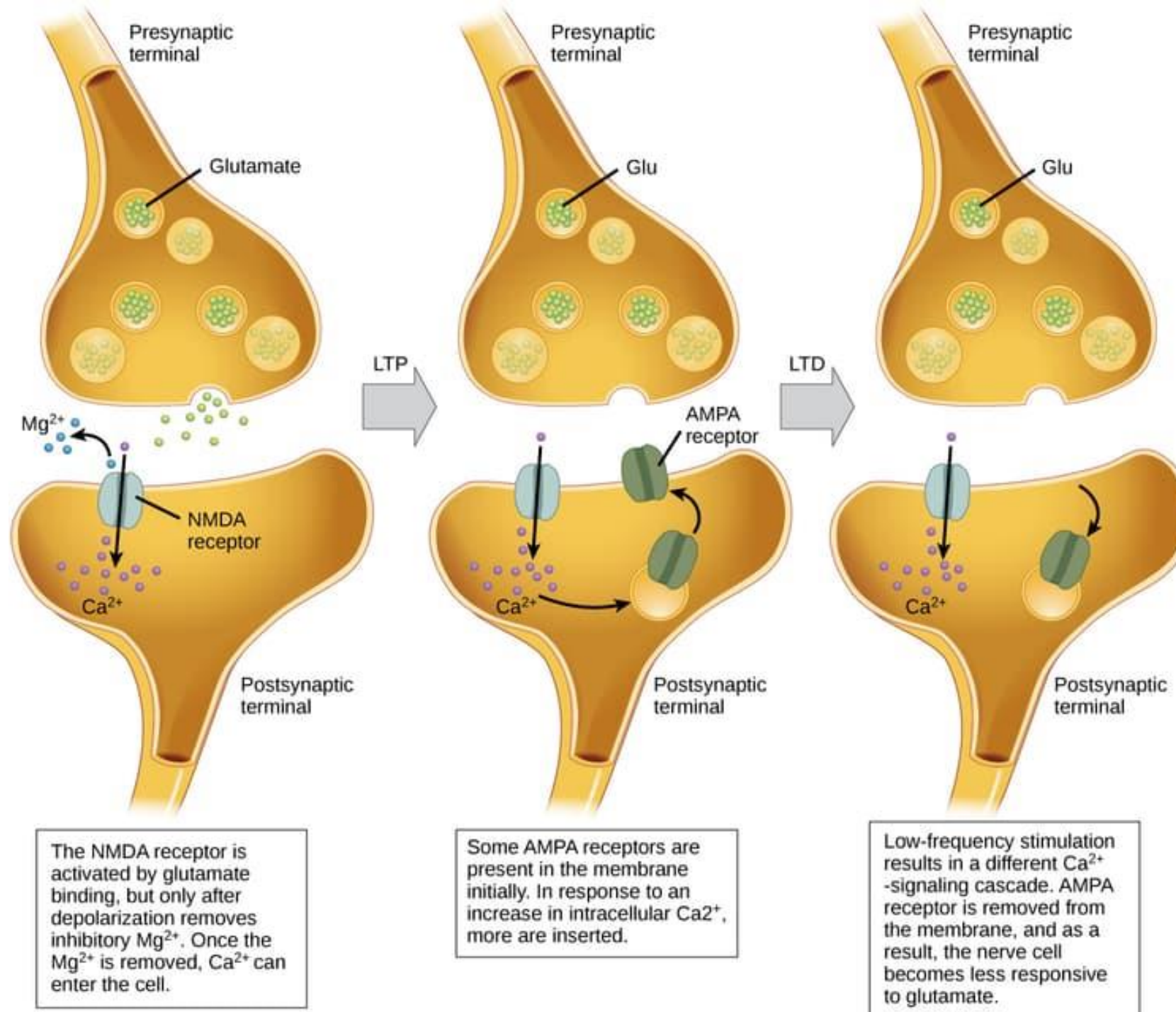
synaptic plasticity

Long-Term Potentiation (LTP)

== A **persistent strengthening** of synapses due to high-frequency stimulation, leading to **increased synaptic efficacy**.

Long-Term Depression (LTD)

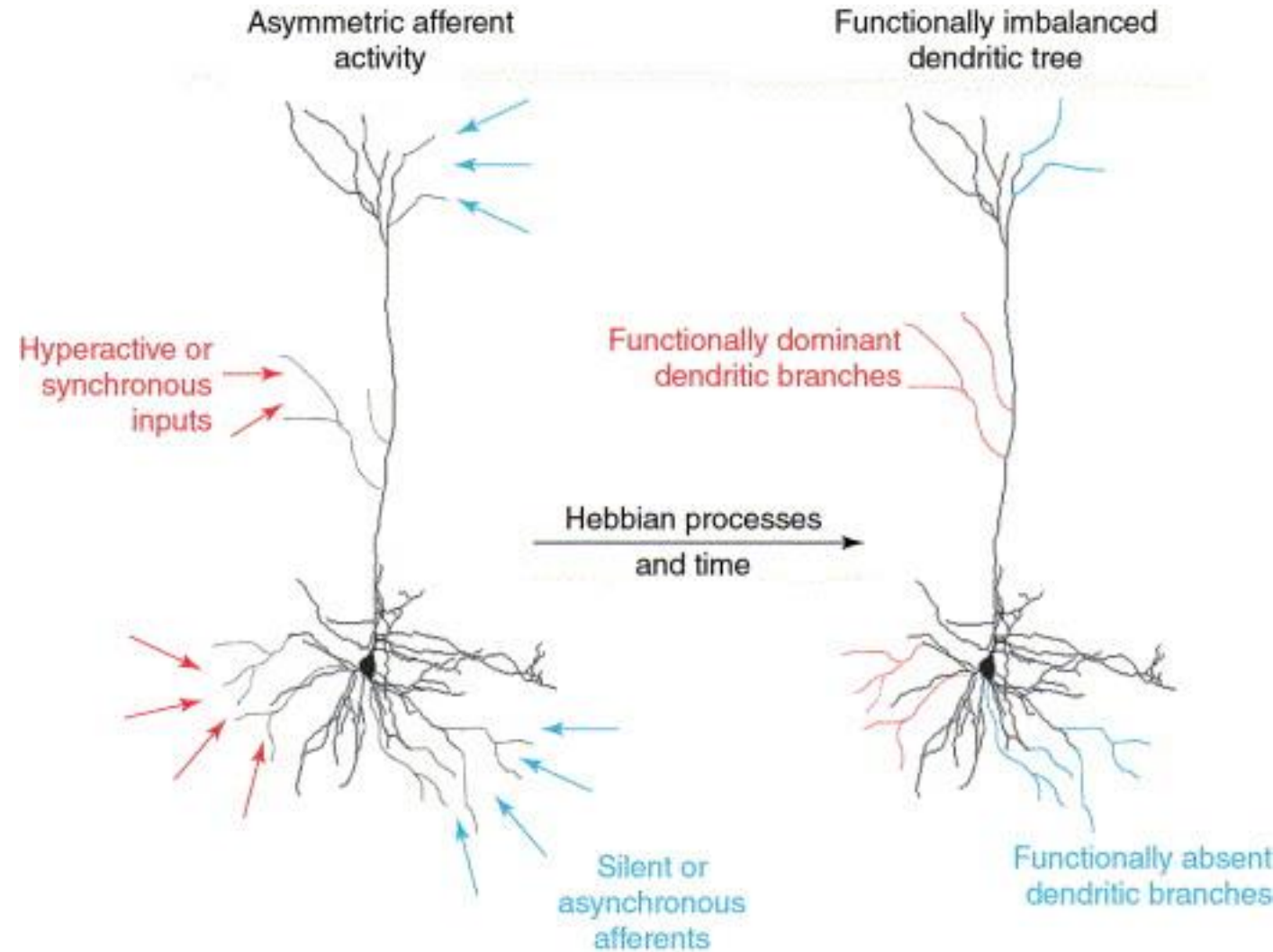
== A **persistent weakening** of synapses due to low-frequency stimulation, leading to reduced synaptic efficacy



What is Hebb's postulate?

What is Hebb's postulate?

"cells that fire together, wire together"



What is Hebb's postulate?

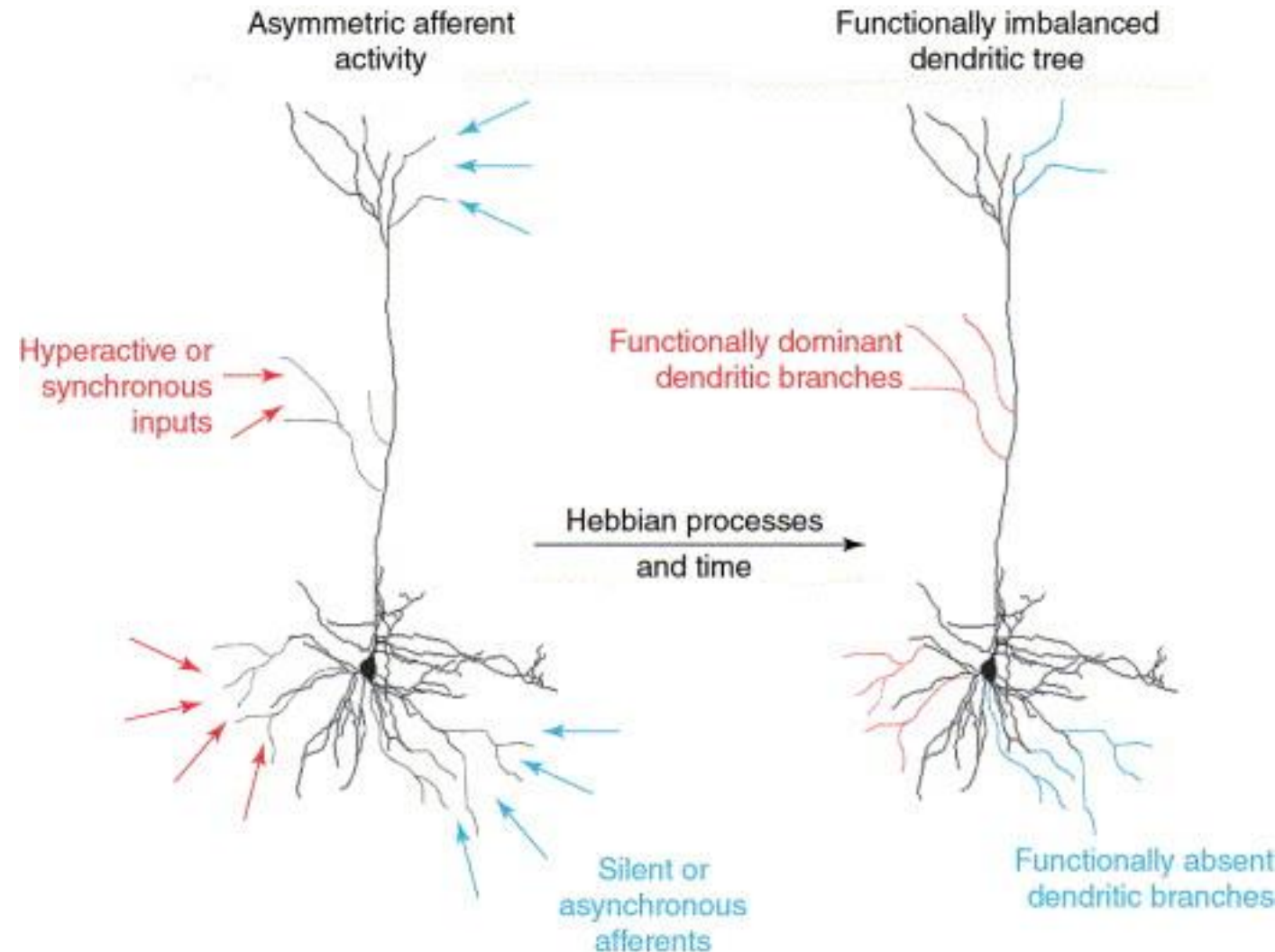
"cells that fire together, wire together"

Synaptic strengthening & LTP:

- When **two neurons repeatedly activate together**, their synaptic connection is strengthened, forming the basis of **long-term potentiation (LTP)**. This mechanism is crucial for associative learning and memory storage, particularly in the hippocampus.

Cell assemblies & memory networks:

- Hebb proposed that memories are stored in **distributed neural networks == cell assemblies**.
- Activation of part of the assembly can reactivate the whole network, enabling memory retrieval.



What is Hebb's postulate?

"cells that fire together, wire together"

Stages of memory

Sensory

ICONIC

ECHOIC

HAPTIC

NOCICEPTIVE

...

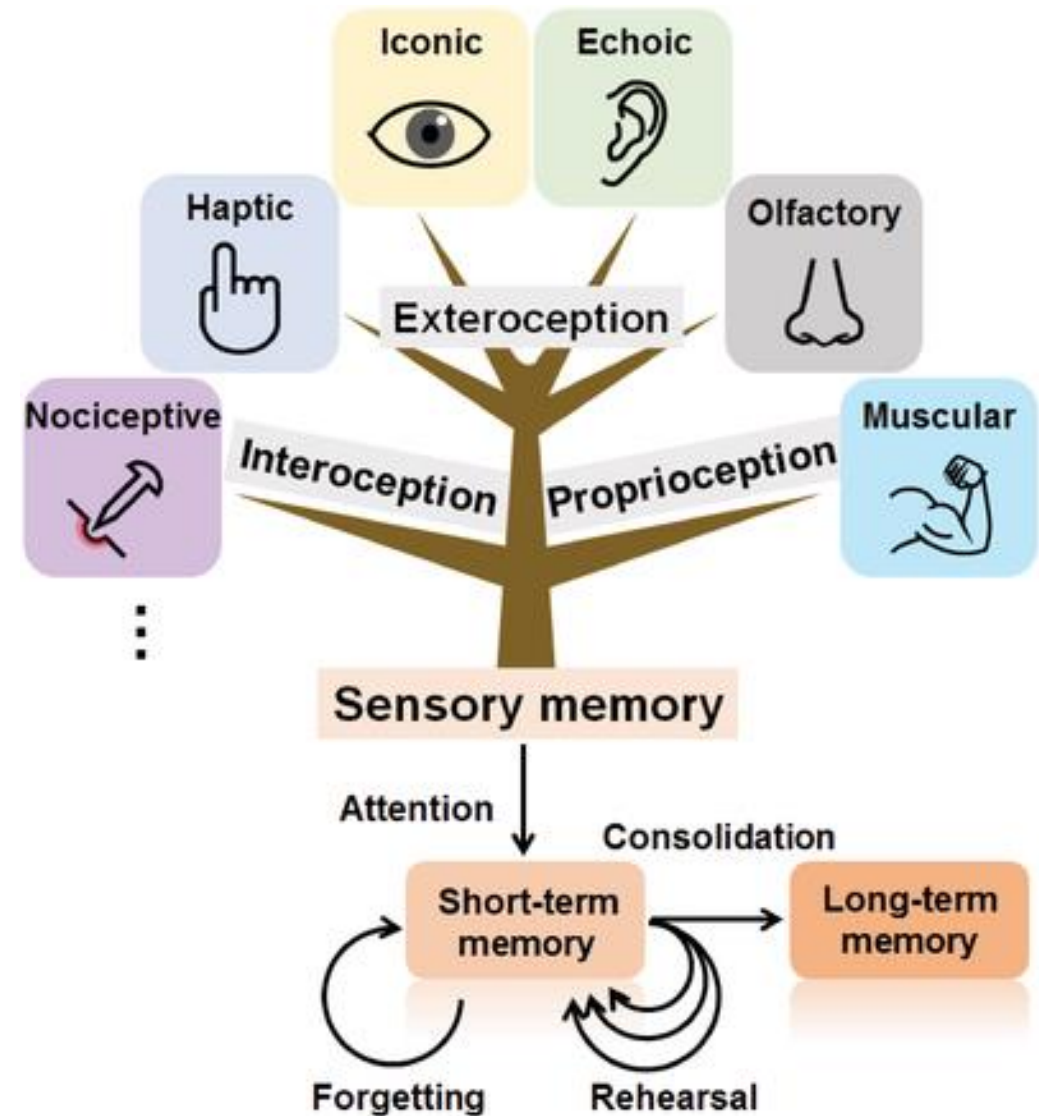
Short-term

Long-term

Stages of memory

Sensory

ICONIC
ECHOIC
HAPTIC
NOCICEPTIVE
...



(visual) sensory memory -- can be experimentally studied by using the partial report technique (Coltheart, 1980)

Memory of faces



Memory of faces

Prosopagnosia

Face Blindness (Prosopagnosia)

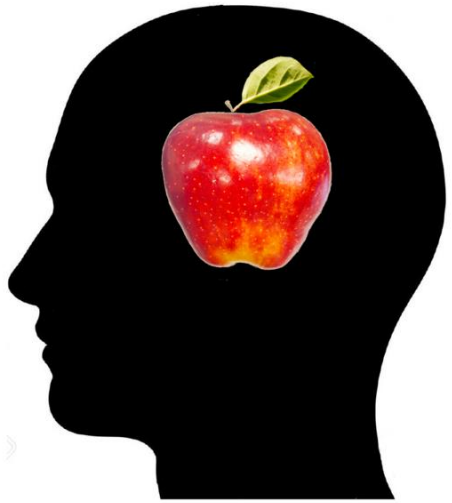


Fusiform face area

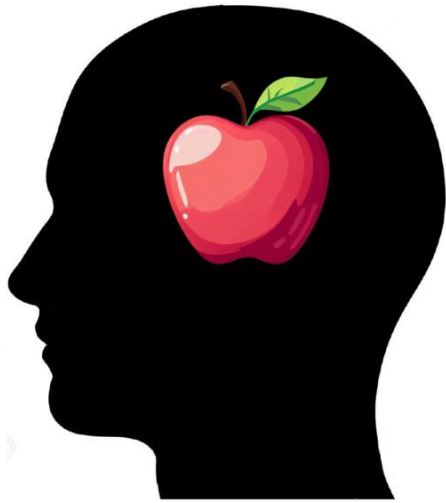
Striate cortex



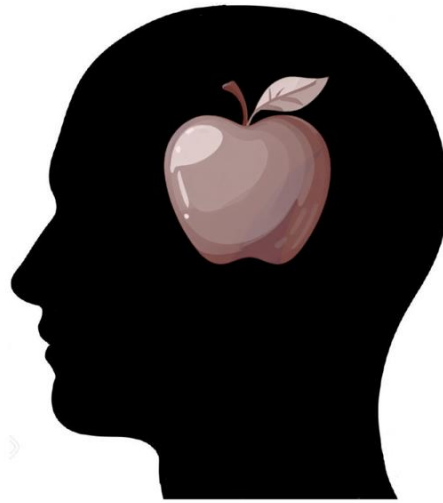
What do we call the ability to picture something?



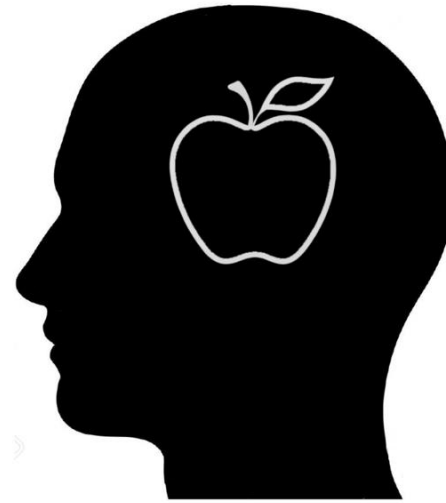
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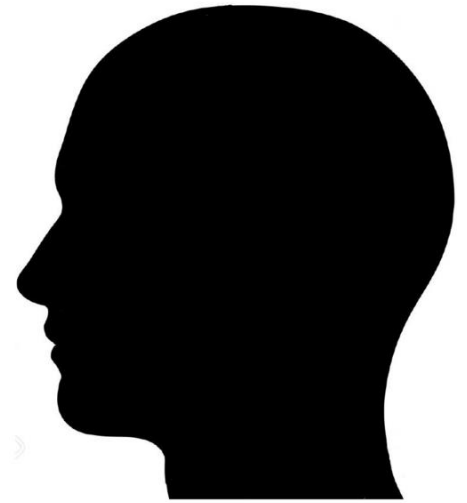
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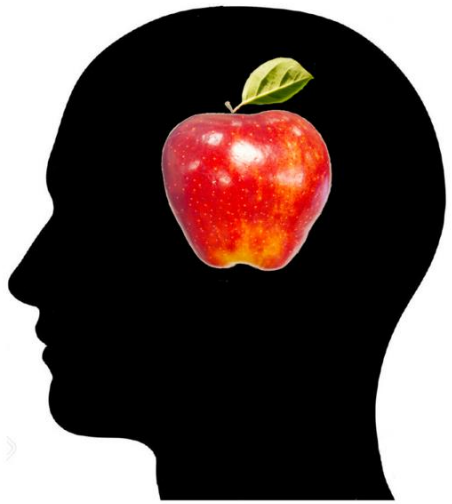


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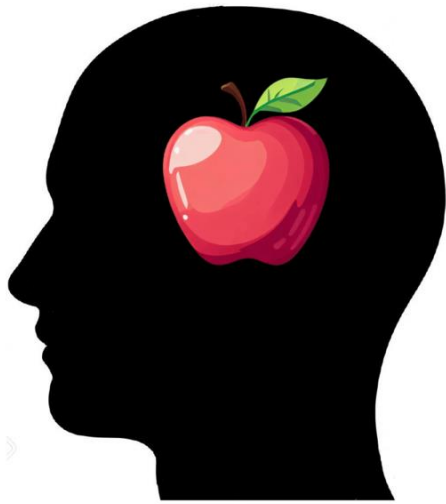


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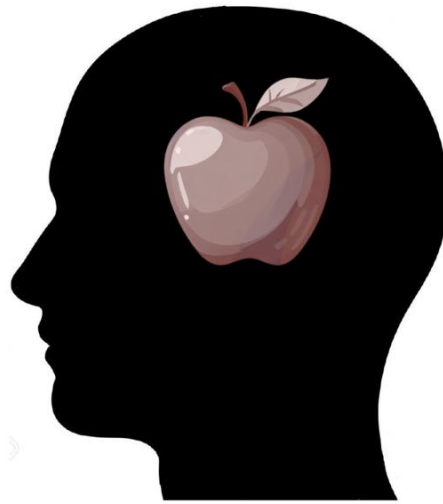
Prosopagnosia $\neq$ **aphantasia**



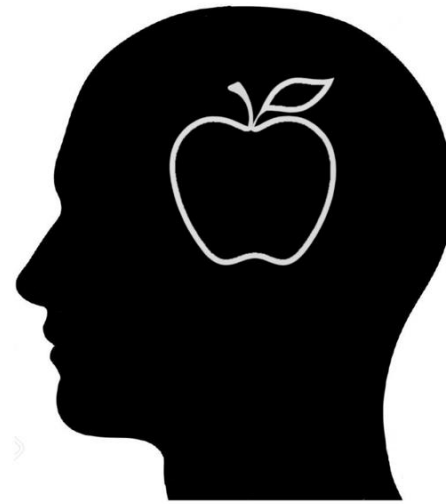
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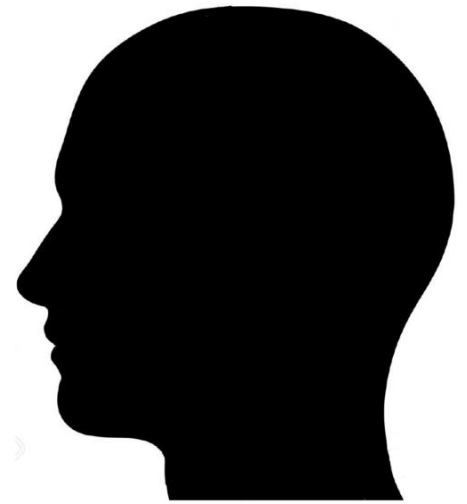
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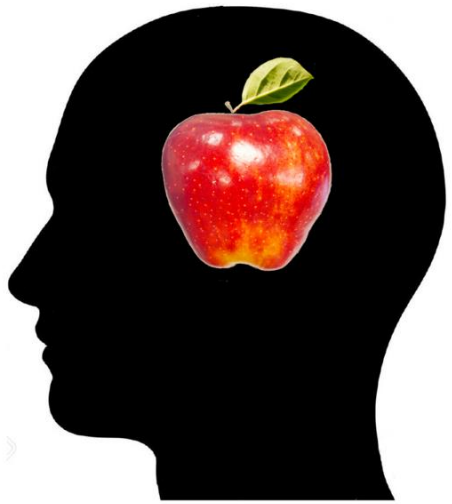


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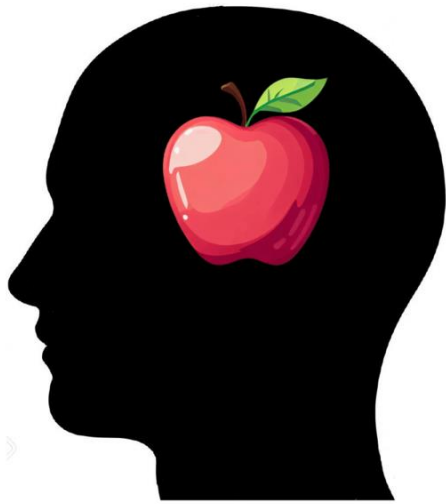


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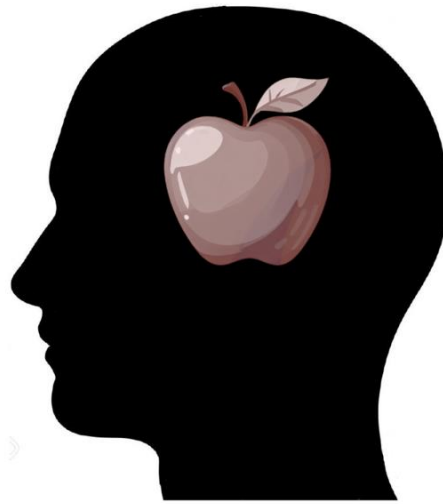
Someone claims they have aphantasia. You are skeptical. How would you go about testing this?



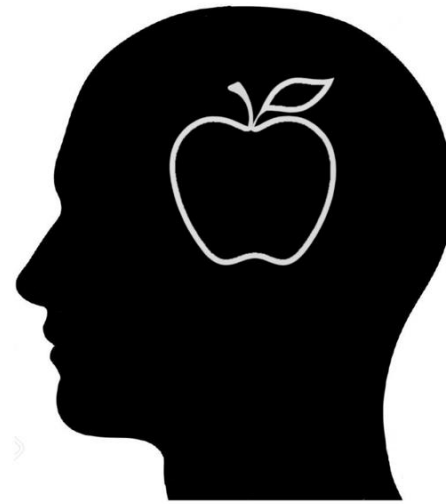
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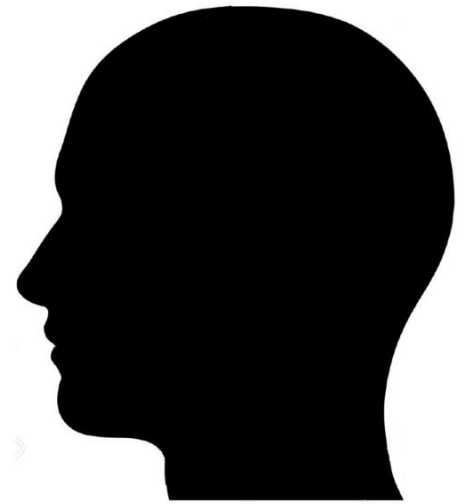
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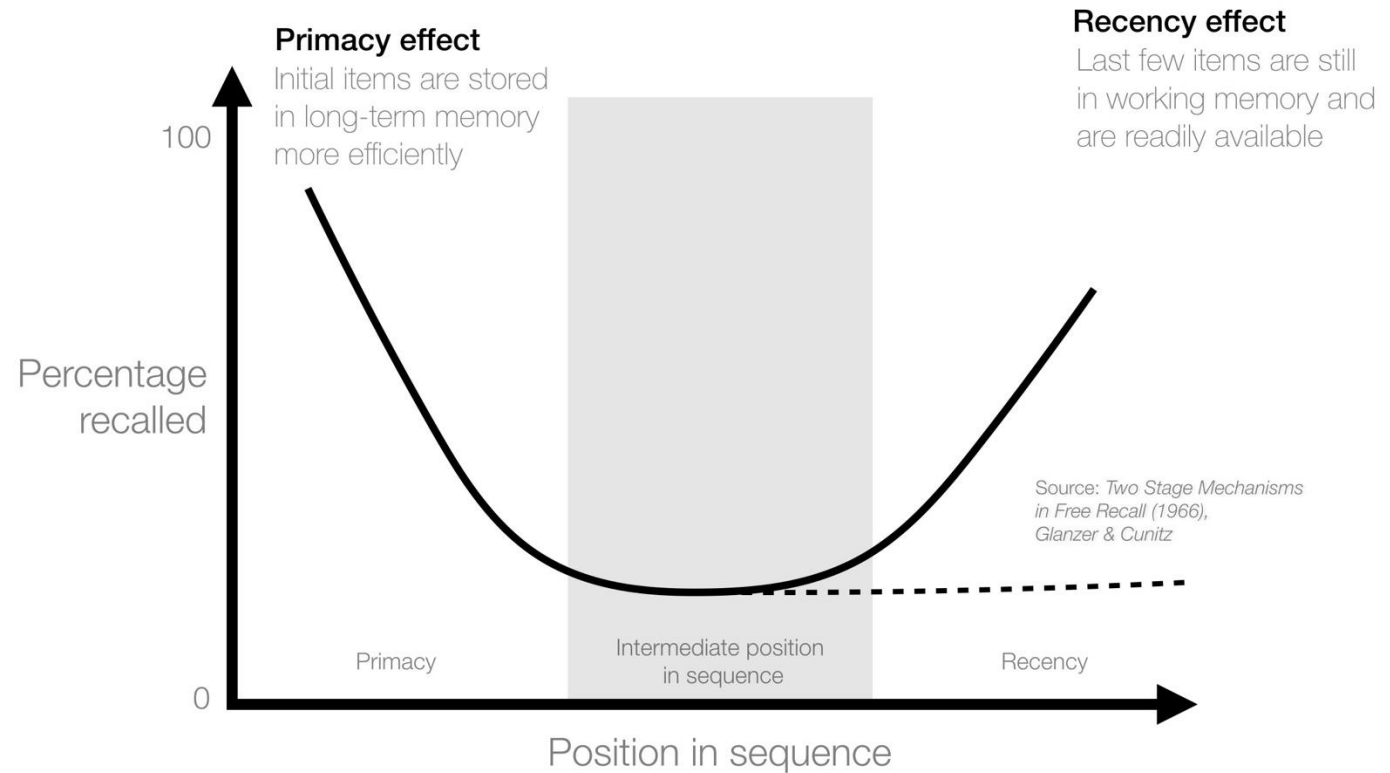
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Sensory memory

Sensory memory: milliseconds to seconds

Short-term memory: seconds to minutes

- Reason for forgetting: decay v. interference (usu interference)
- *Order* matters: serial pattern effect
 - **Primacy and Recency effects** – we are better at remembering things in the beginning and ending of a list
 - **Primacy:** transfer occurs from short to long-term memory through repetition
 - **Recency:** retention in short-term memory



Sensory memory

Attention

Sensory Inputs → Sensory Register → Short-term Storage
→ Long-term storage

Rehearsal

At any stage, information can be lost due to decay or interference,
or both

Craik and Lockhart (1972): level of processing matters

Episodic memory

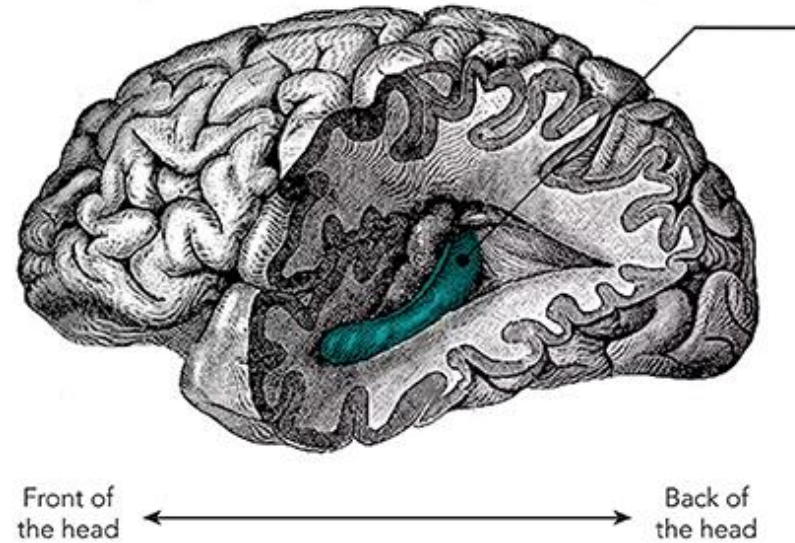
Q: what is episodic memory?

Episodic memory

Q: Where is episodic memory? What part of the brain localizes it?

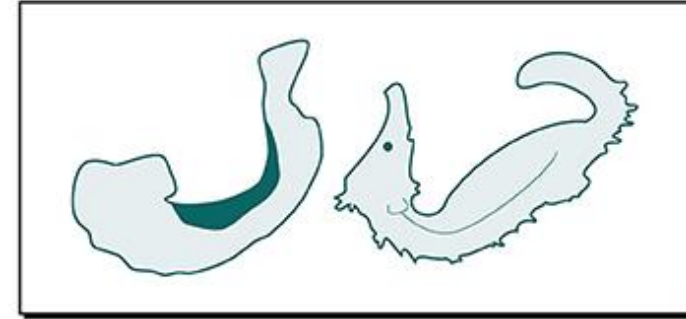
Episodic memory

hippocampus



Hippocampus

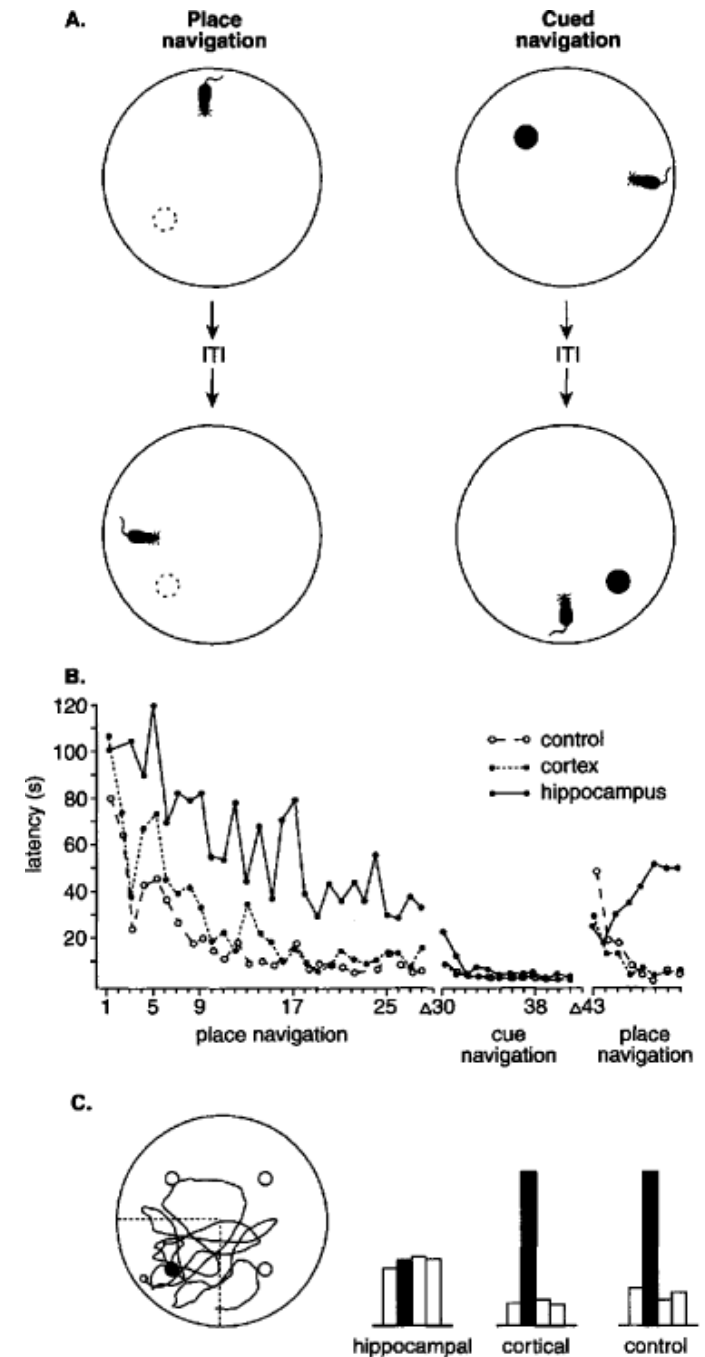
A brain region that is important for supporting episodic memory



- **Episodic memory encoding** occurs primarily in the **medial temporal lobe**, especially the **hippocampus**, **entorhinal cortex**, and **perirhinal cortex**.
- The **prefrontal cortex** plays a role in retrieval, organisation, and contextual recall.
- **Hippocampal damage** (e.g., **patient HM**) leads to anterograde amnesia, but remote memories may be spared due to *systems consolidation* (transfer to neocortex).
- Some amnesic patients (e.g., HM, KC) **retain remote episodic and semantic memories** because older memories rely less on the hippocampus over time and become cortically distributed.

Animal models of memory

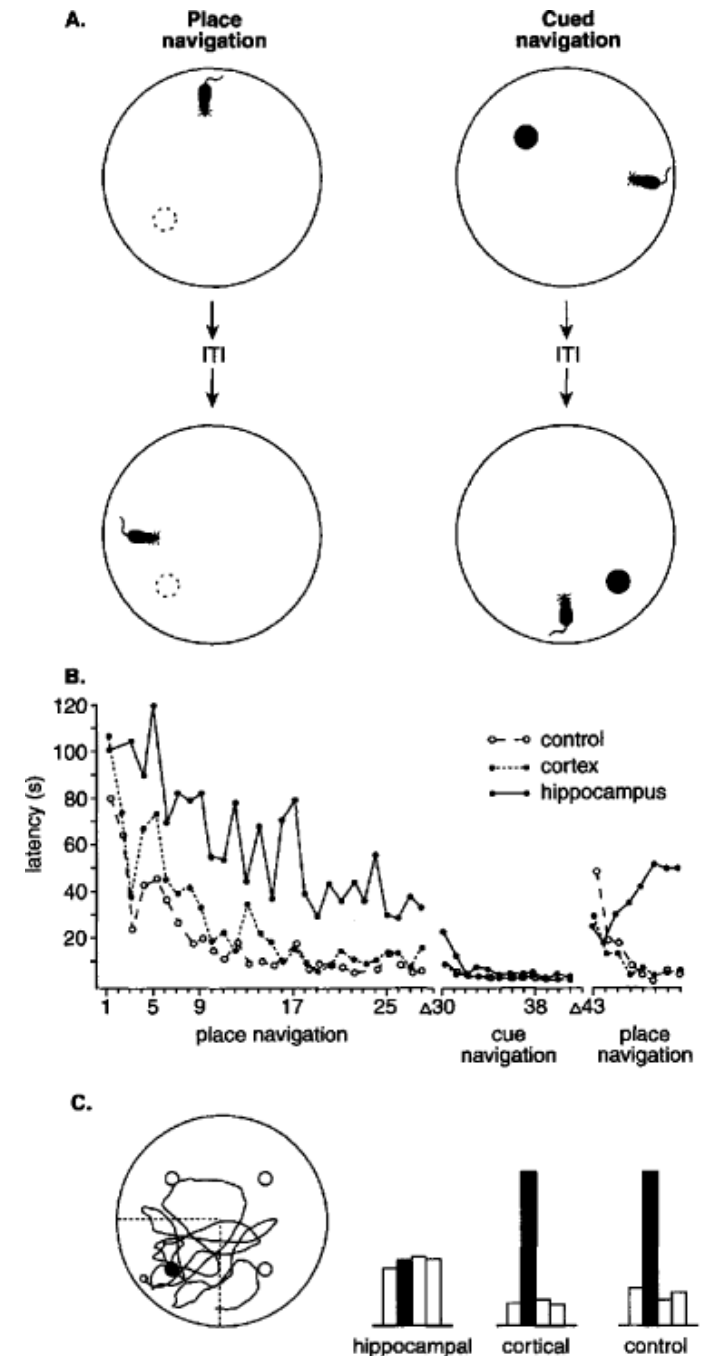
(Morris et al., 1982)



Animal models of memory

(Morris et al., 1982)

- Role of **place cells** (O'Keefe, Nobel Prize 2014) and **grid cells** (Moser & Moser) in spatial memory
- Synaptic plasticity: **Long-Term Potentiation (LTP)** in hippocampus (NMDA receptor activation, AMPA receptor insertion)

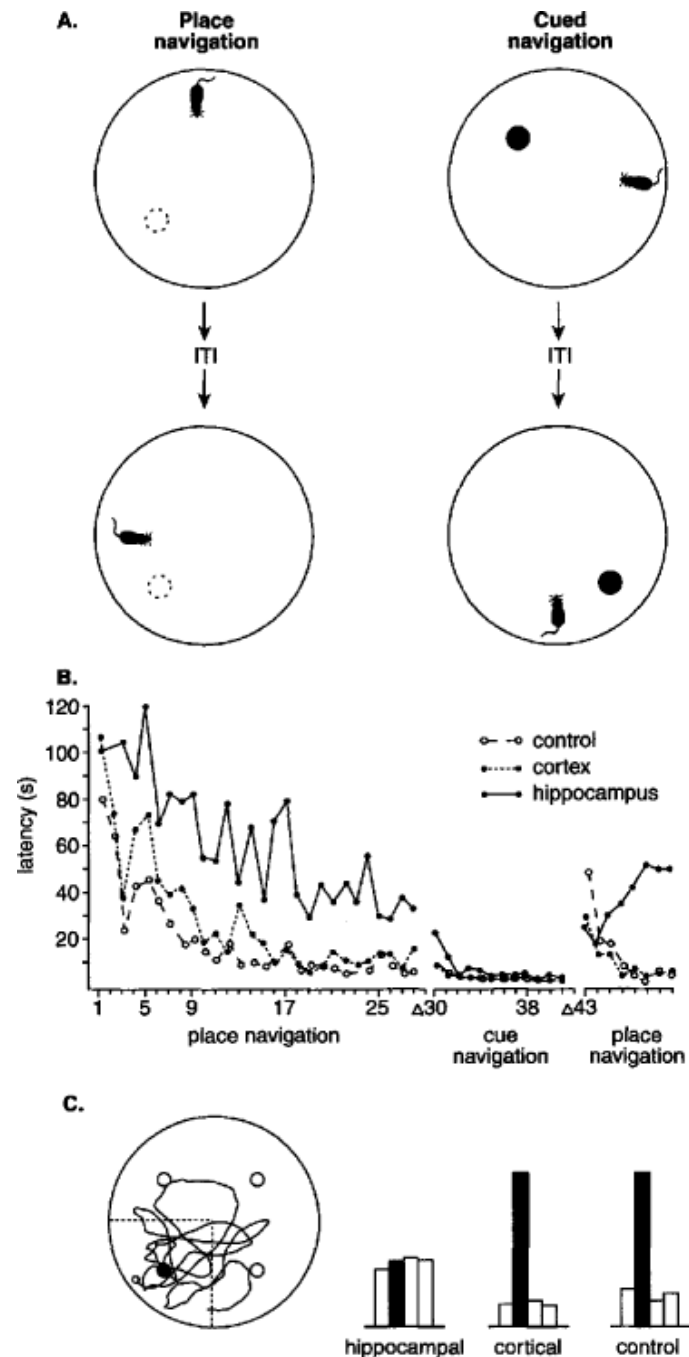


(Morris et al., 1982)

Animal models of memory

- Role of **place cells** (O'Keefe, Nobel Prize 2014) and **grid cells** (Moser & Moser) in spatial memory

Q: where are the place cells and grid cells found?



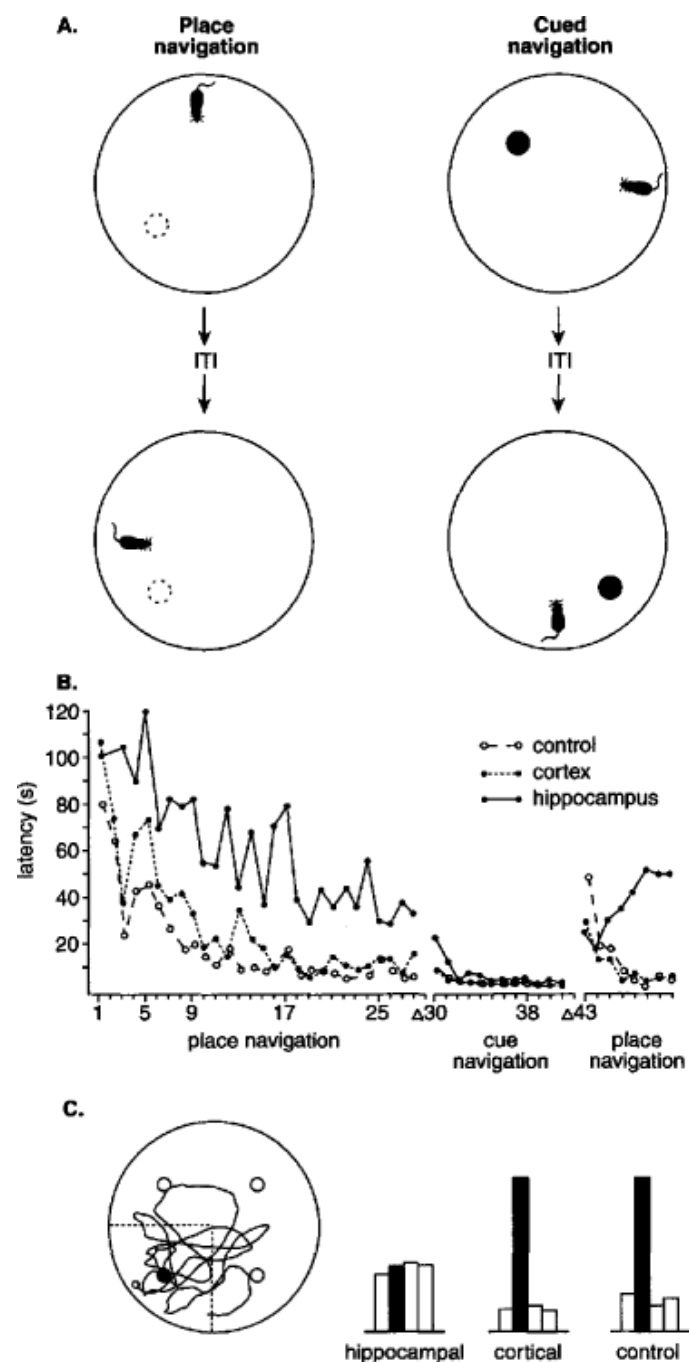
(Morris et al., 1982)

Animal models of memory

Spatial memory: place cells and grid cells

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Q: where are the place cells and grid cells found?



Animal models of memory

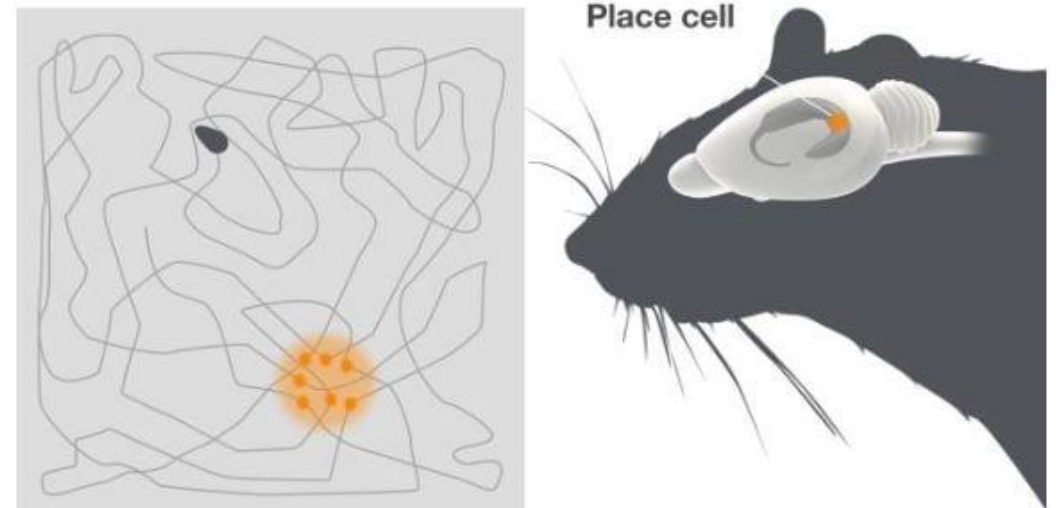
Spatial memory: place cells and grid cells

- The **hippocampus** plays a crucial role in various stages of learning, including memory encoding, consolidation, and recall, particularly in the context of spatial learning.

- **Place cells** in the hippocampus exhibit **distinct firing patterns in different locations** within an environment..

... forming a **cognitive representation of the explored space**. When learning goals within the environment, these place cells shift their firing patterns to align with the learned

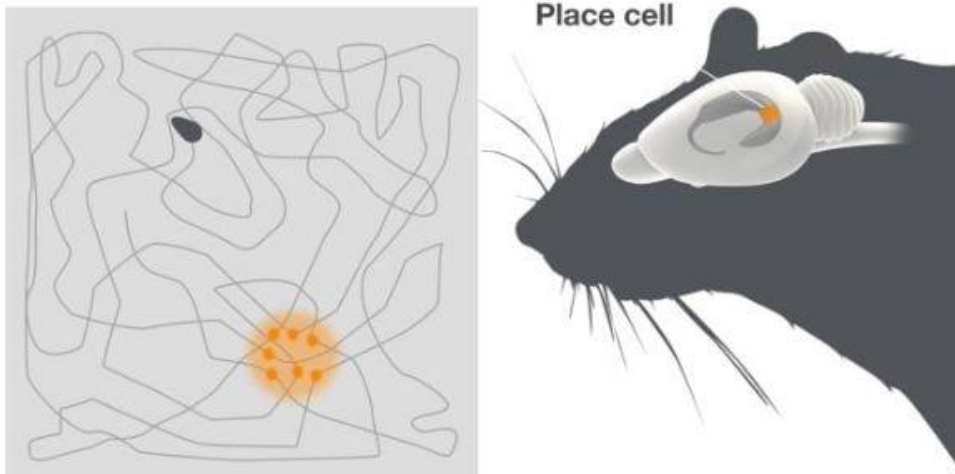
- **Place Cells:** Located in the hippocampus (especially in the CA1 and CA3 subfields).
- **Grid Cells:** Found in the entorhinal cortex (medial entorhinal cortex - MEC), which is the main input to the hippocampus



Animal models of memory

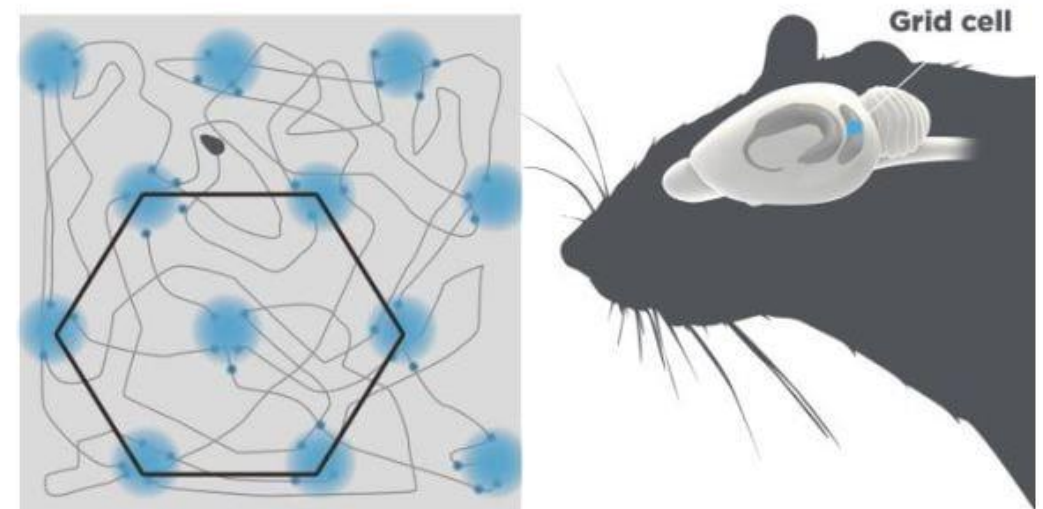
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*2014 Nobel Prize in Physiology
(O'Keefe, Moser)*

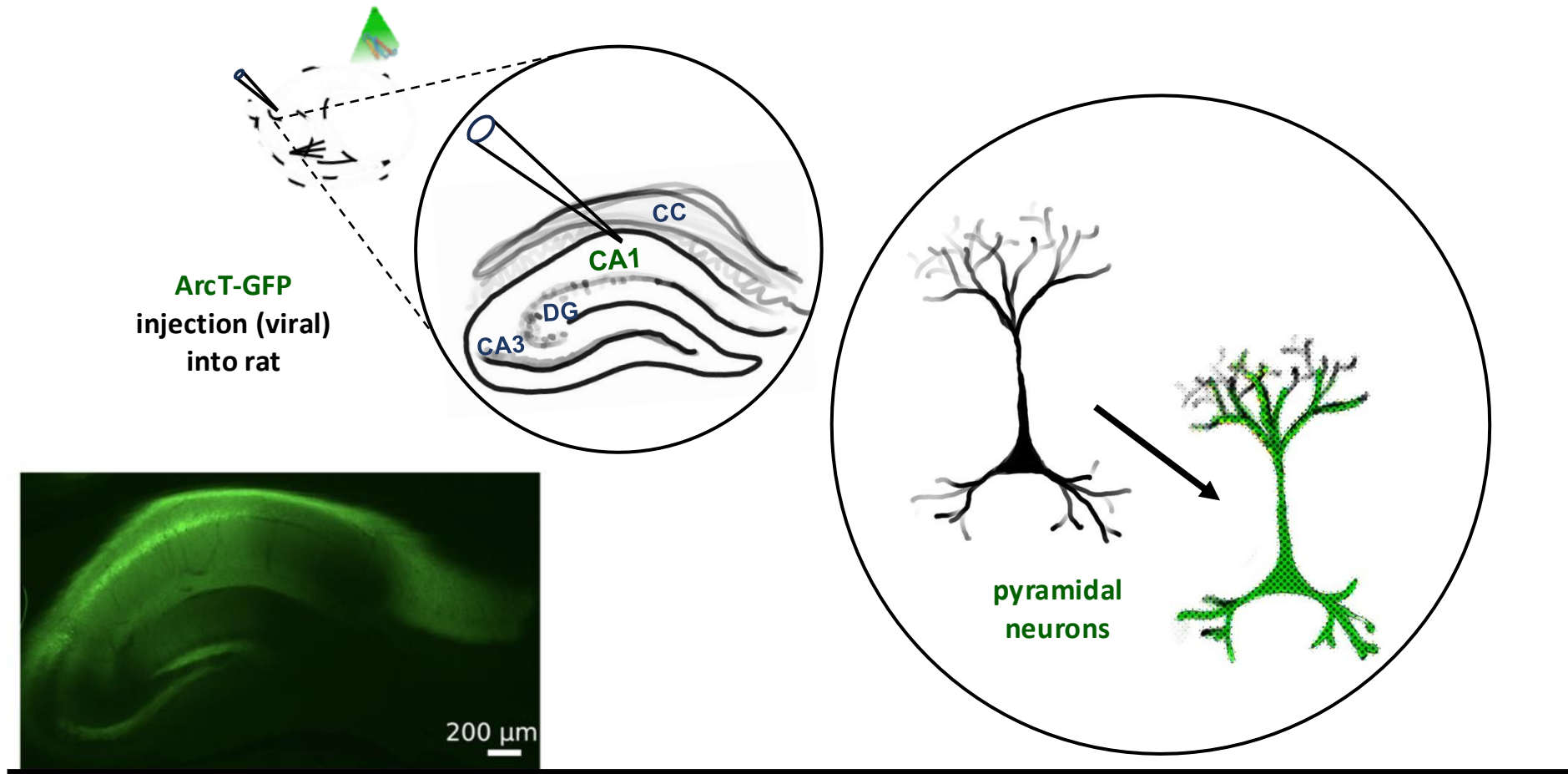
*Major input to hippocampus =
entorhinal cortex*



Animal models of memory

Methods

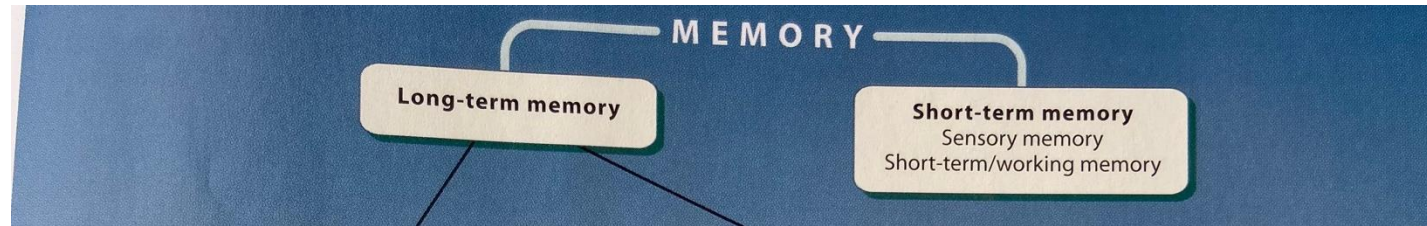
- These rats had surgery, injected with a virus to express **Archaeorhodopsin-GFP** in the CA1
- Rats then had electrodes positioned in **CA1 (hippocampus)**
- this surgical process aimed to allow **real-time decoding of neuronal activity and optogenetic stimulation**



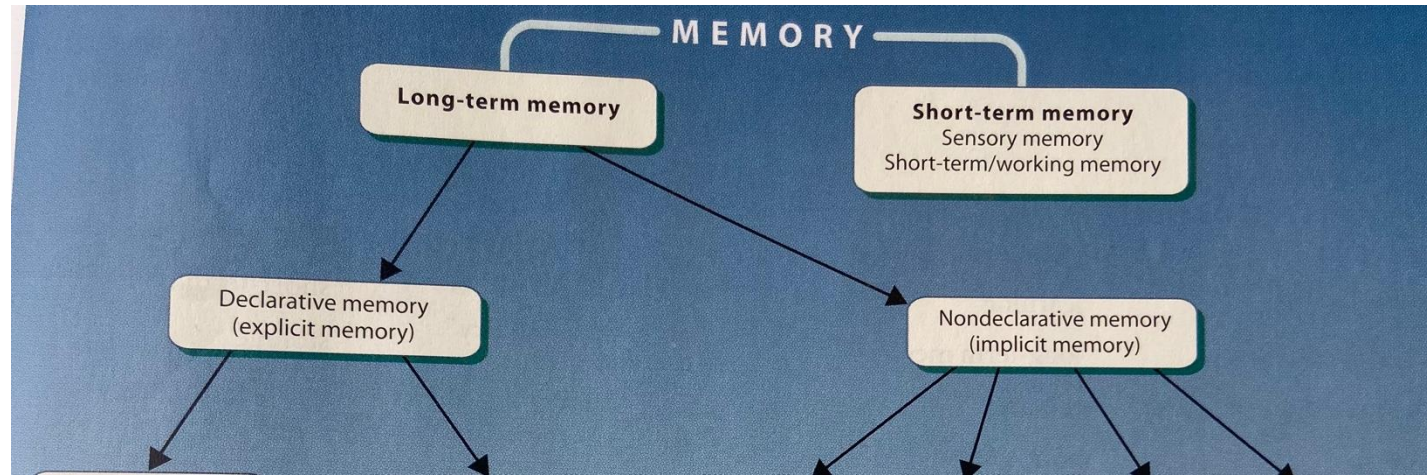
Amnesia

- **Medial Temporal Lobe** – pt H.M. and R.B.
Mainly anterograde amnesia, but some retrograde amnesia
- **Mamillary bodies** – Korsakoff's Syndrome
Anterograde and retrograde amnesia a/w alcoholism
- **Learning in amnesia** – are episodic, semantic, and procedural information different?

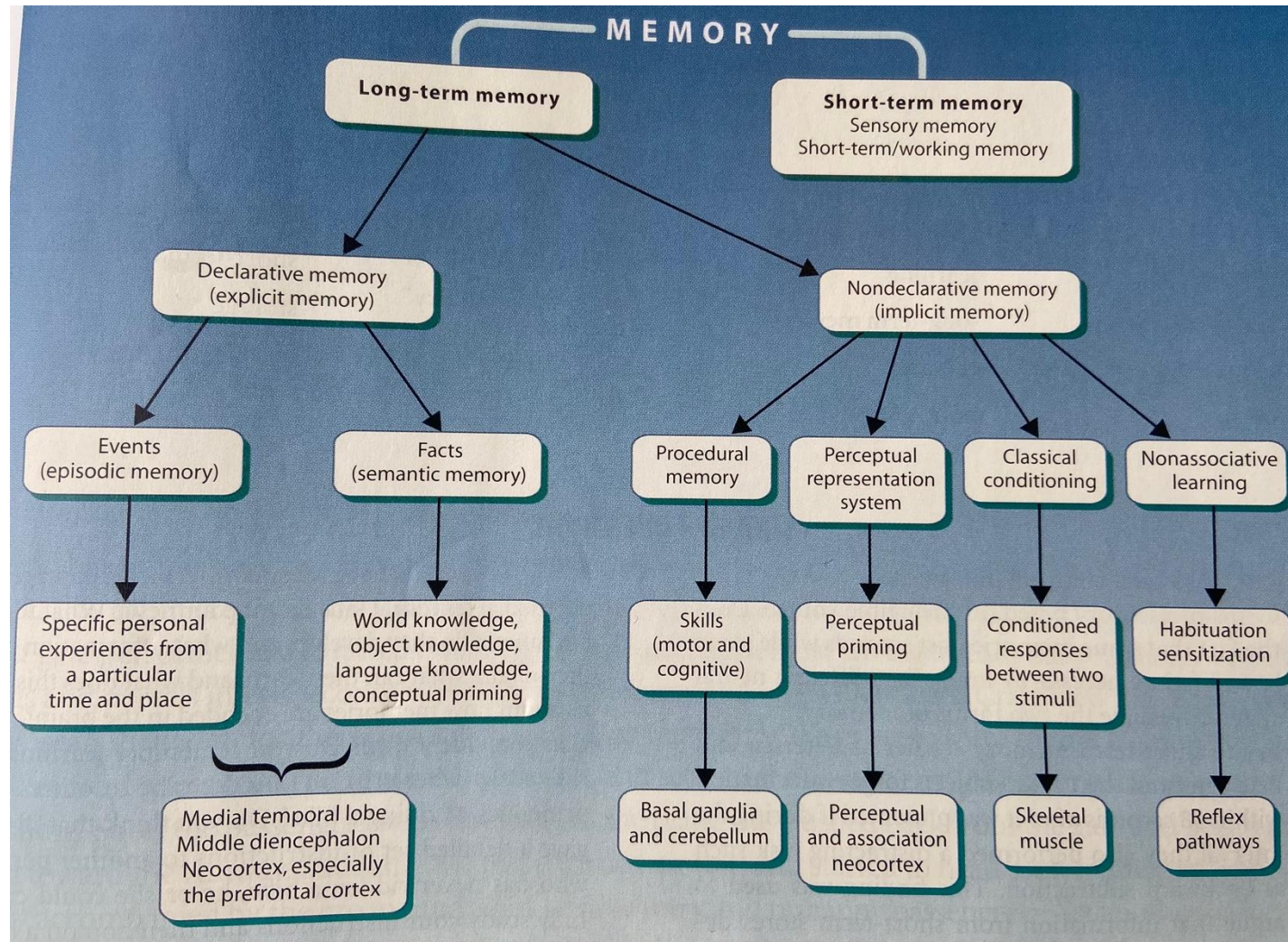
Different types of memory: compartmentalization



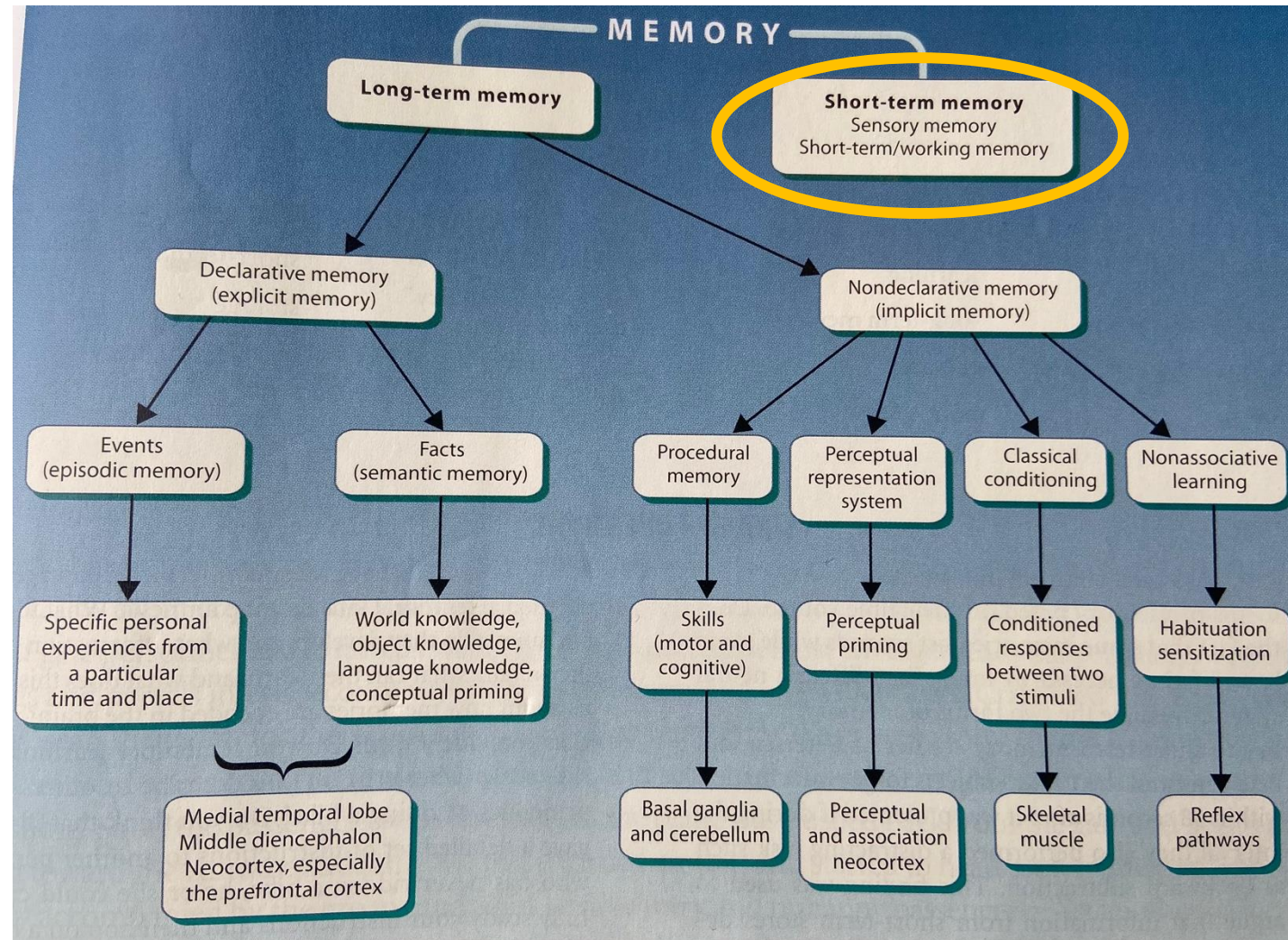
Different types of memory



Different types of memory



What is working memory?



Working memory

1970s..

- Baddeley et al proposed the first variant of working memory – information that can be acted on and processed
- Somehow, this will lead to long-term memory if retained
- Pts with amnesia help explain how this works
Shallice and Warrington – pt w/ L perisylvian damaged reduced digit span to 2, but could make long-term memories

Working memory

Phonological loop (left supramarginal gyrus, and L premotor region)

Visuospatial sketchpad (parieto-occipital region of both hemispheres)

- Baddeley et al proposed the first variant of working memory – information that can be acted on and processed

Baddeley & Hitch (1974) Working Memory Model

- Central Executive: Attention control, task-switching (prefrontal cortex)
- Phonological Loop: Verbal STM (left temporoparietal regions)
- Visuospatial Sketchpad: Spatial memory (occipital & parietal lobes)
- Episodic Buffer: Integration with LTM

Experimental Evidence

- Dual-task studies (e.g., articulatory suppression)
- n-back task measuring working memory capacity
- Neuroimaging: Activation in **dorsolateral prefrontal cortex**

Cognitive ability

Measurement

‘Global’ Cognitive Ability

MoCA (Montreal Cognitive Assessment)

Famously used by this guy



as evidence of cognitive prowess...

Cognitive ability

Measurement

‘Global’ Cognitive Ability

Donald Trump has “incredible genes” and could have lived to 200 years old if his diet had been better in recent years, the White House presidential physician said.

Dr Ronny Jackson, who performed Mr Trump's first medical exam as president, declared him “fit for duty” physically and mentally.

As well as a thorough physical, the US president asked to be administered a mental test and was given the Montreal Cognitive Assessment. Dr Jackson performed the “difficult” cognitive test and Mr Trump scored 30 out of 30.

Cognitive ability

Measurement

‘Global’ Cognitive Ability

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NAME :
Education :
Sex :
Date of birth :
DATE :

5

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End

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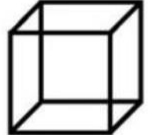
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[]

Copy
cube



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Draw CLOCK (Ten past eleven)
(3 points)

[]

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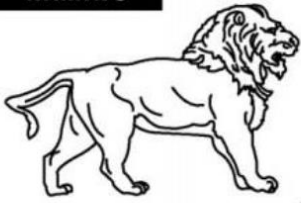
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Contour

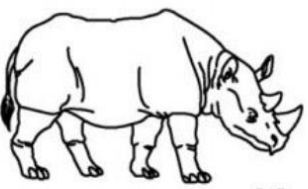
Numbers

Hands

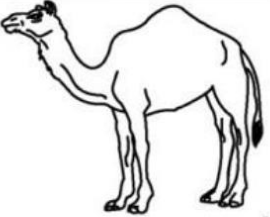
___/5



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___/3

MEMORY

Read list of words, subject must repeat them. Do 2 trials. Do a recall after 5 minutes.

FACE

VELVET

CHURCH

DAISY

RED

1st trial

2nd trial

No points

ATTENTION

Read list of digits (1 digit/ sec.). Subject has to repeat them in the forward order [] 2 1 8 5 4
Subject has to repeat them in the backward order [] 7 4 2

Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors
[] FBACMNAAJKLBFAFAKDEAAAJAMOF AAB

Serial 7 subtraction starting at 100 [] 93 [] 86 [] 79 [] 72 [] 65
4 or 5 correct subtractions: 3 pts, 2 or 3 correct: 2 pts, 1 correct: 1 pt, 0 correct: 0 pt

___/2

___/1

___/3

LANGUAGE

Repeat : I only know that John is the one to help today. []
The cat always hid under the couch when dogs were in the room. []

Fluency / Name maximum number of words in one minute that begin with the letter F [] ____ (N ≥ 11 words)

___/2

___/1

ABSTRACTION

Similarity between e.g. banana - orange = fruit [] train - bicycle [] watch - ruler

___/2

DELATED RECALL

Has to recall words WITH NO CUE

FACE

VELVET

CHURCH

DAISY

RED

Points for UNCUE recall only

___/5

Optional

Category cue

Multiple choice cue

ORIENTATION

[] Date [] Month [] Year [] Day [] Place [] City

___/6

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Normal ≥ 26 / 30

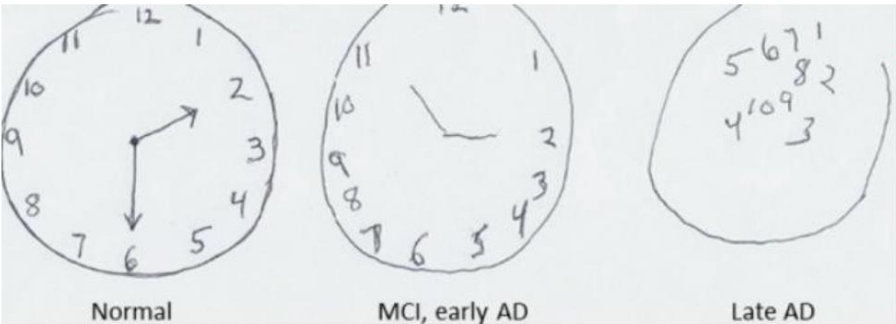
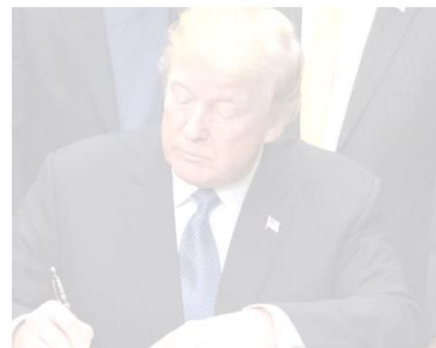
TOTAL ___/30

Add 1 point if ≤ 12 yr edu

‘Global’ Cognitive Ability

MoCA (Montreal Cognitive Assessment)

Famously used by this guy



as evidence of cognitive prowess...

Alzheimer’s patients may omit numbers, misplace hands, draw a distorted shape, or fail to complete the clock entirely— why?

Dr Ronny Jackson, who performed Mr Trump’s first medical exam as president, declared him “fit for duty” physically and mentally.

As well as a thorough physical, the US president asked to be administered a mental test and was given the Montreal Cognitive Assessment. Dr Jackson performed the “difficult” cognitive test and Mr Trump scored 30 out of 30.

MONTREAL COGNITIVE ASSESSMENT (MOCA)

NAME : _____ Education : _____ Date of birth : _____
Sex : _____ DATE : _____

VISUOSPATIAL / EXECUTIVE

Copy cube  **Draw CLOCK (Ten past eleven) (3 points)**

Points: ☐ Contour ☐ Numbers ☐ Hands ☐ /5

MEMORY

Read list of words, subject must repeat them. Do 2 trials. Do a recall after 5 minutes.

	FACE	VELVET	CHURCH	DAISY	RED
1st trial					
2nd trial					

No points

ATTENTION

Read list of digits (1 digit/ sec.). Subject has to repeat them in the forward order ☐ 2 1 8 5 4
Subject has to repeat them in the backward order ☐ 7 4 2

Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors
☐ F B A C M N A A J K L B A F A K D E A A A J A M O F A A B

Serial 7 subtraction starting at 100 ☐ 93 ☐ 86 ☐ 79 ☐ 72 ☐ 65

4 or 5 correct subtractions: 3 pts, 2 or 3 correct: 2 pts, 1 correct: 1 pt, 0 correct: 0 pt

LANGUAGE

Repeat : I only know that John is the one to help today. ☐
The cat always hid under the couch when dogs were in the room. ☐

Fluency / Name maximum number of words in one minute that begin with the letter F ☐ (N ≥ 11 words)

ABSTRACTION

Similarity between e.g. banana - orange = fruit ☐ train - bicycle ☐ watch - ruler

DELAYED RECALL

	FACE	VELVET	CHURCH	DAISY	RED
WITH NO CUE	☐	☐	☐	☐	☐
Category cue					
Multiple choice cue					

Points for UNCUED recall only

ORIENTATION

☐ Date ☐ Month ☐ Year ☐ Day ☐ Place ☐ City

© Z.Nosreddine MD Version 7.0 www.mocatest.org Normal ≥ 26 / 30 TOTAL ☐ /30

Administered by: _____ Add 1 point if ≤ 12 yr edu

Q: Other than the MoCA, can you think of another widely used cognitive test used by clinicians?

‘Global’ Cognitive Ability

MMSE – Mini-Mental State Examination

Mini-Mental State Examination (MMSE)

Patient's Name: _____ Date: _____

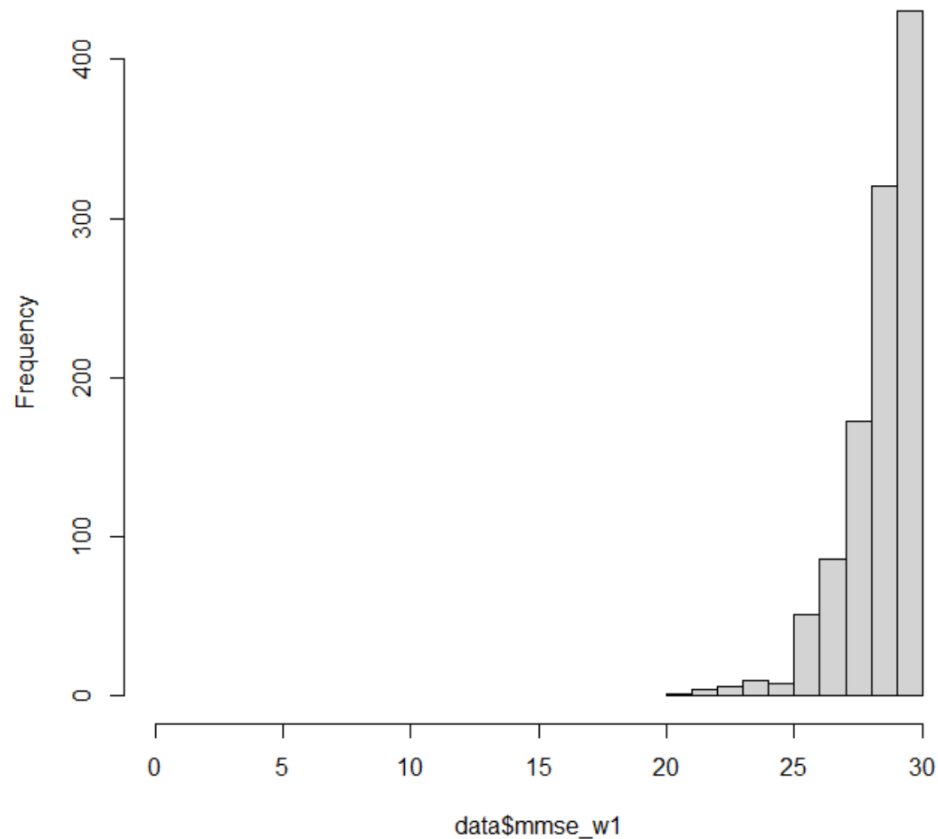
Instructions: Ask the questions in the order listed. Score one point for each correct response within each question or activity.

Maximum Score	Patient's Score	Questions
5		"What is the year? Season? Date? Day of the week? Month?"
5		"Where are we now: State? County? Town/city? Hospital? Floor?"
3		The examiner names three unrelated objects clearly and slowly, then asks the patient to name all three of them. The patient's response is used for scoring. The examiner repeats them until patient learns all of them, if possible. Number of trials: _____
5		"I would like you to count backward from 100 by sevens." (93, 86, 79, 72, 65, ...) Stop after five answers. Alternative: "Spell WORLD backwards." (D-L-R-O-W)
3		"Earlier I told you the names of three things. Can you tell me what those were?"
2		Show the patient two simple objects, such as a wristwatch and a pencil, and ask the patient to name them.
1		"Repeat the phrase: 'No ifs, ands, or buts.'"
3		"Take the paper in your right hand, fold it in half, and put it on the floor." (The examiner gives the patient a piece of blank paper.)
1		"Please read this and do what it says." (Written instruction is "Close your eyes.")
1		"Make up and write a sentence about anything." (This sentence must contain a noun and a verb.)
1		"Please copy this picture." (The examiner gives the patient a blank piece of paper and asks him/her to draw the symbol below. All 10 angles must be present and two must intersect.)
30		TOTAL

A widely-used screener for ‘grading the cognitive state of patients for the clinician’.

‘Global’ Cognitive Ability

MMSE – Mini-Mental State Examination



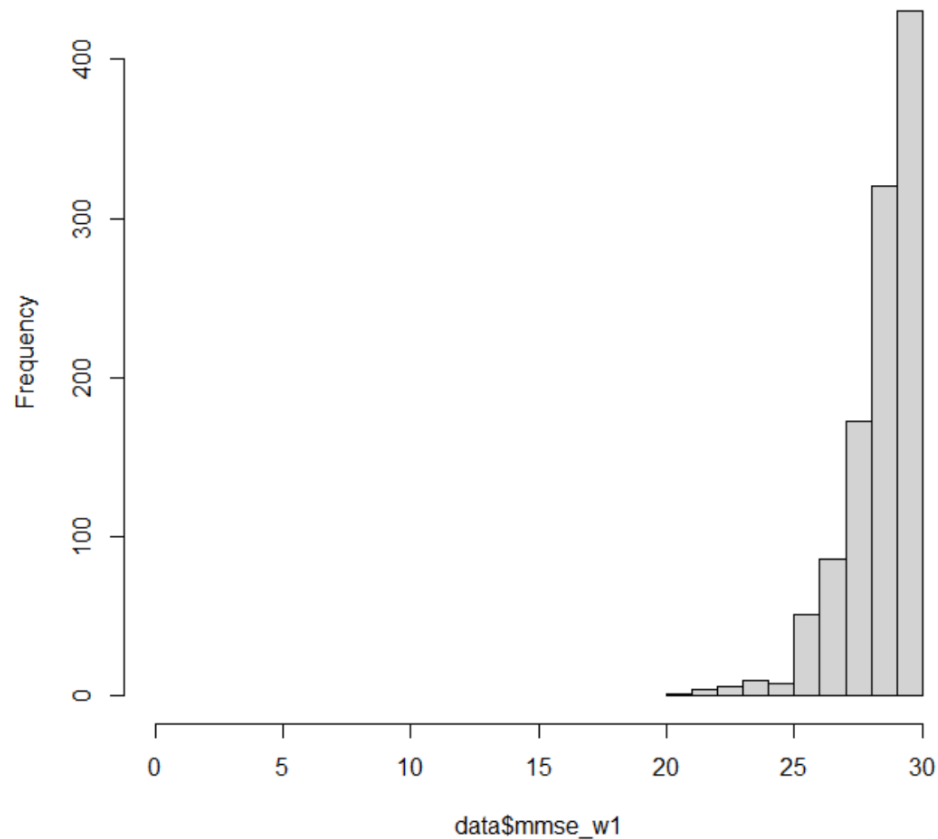
A widely-used screener for ‘grading the cognitive state of patients for the clinician’.

In general these tests (MMSE, MoCA and others) are useful for ascertaining cognitive function among patient groups in clinic and research settings (lesions, dementia etc).

Is this a useful test for assessing memory in ‘healthy’ patients – aka so called ‘normative’ cognitive ageing?

‘Global’ Cognitive Ability

MMSE – Mini-Mental State Examination



A widely-used screener for ‘grading the cognitive state of patients for the clinician’.

In general these tests (MMSE, MoCA and others) are useful for ascertaining cognitive function among patient groups in clinic and research settings (lesions, dementia etc).

Generally not so helpful within the normal range.

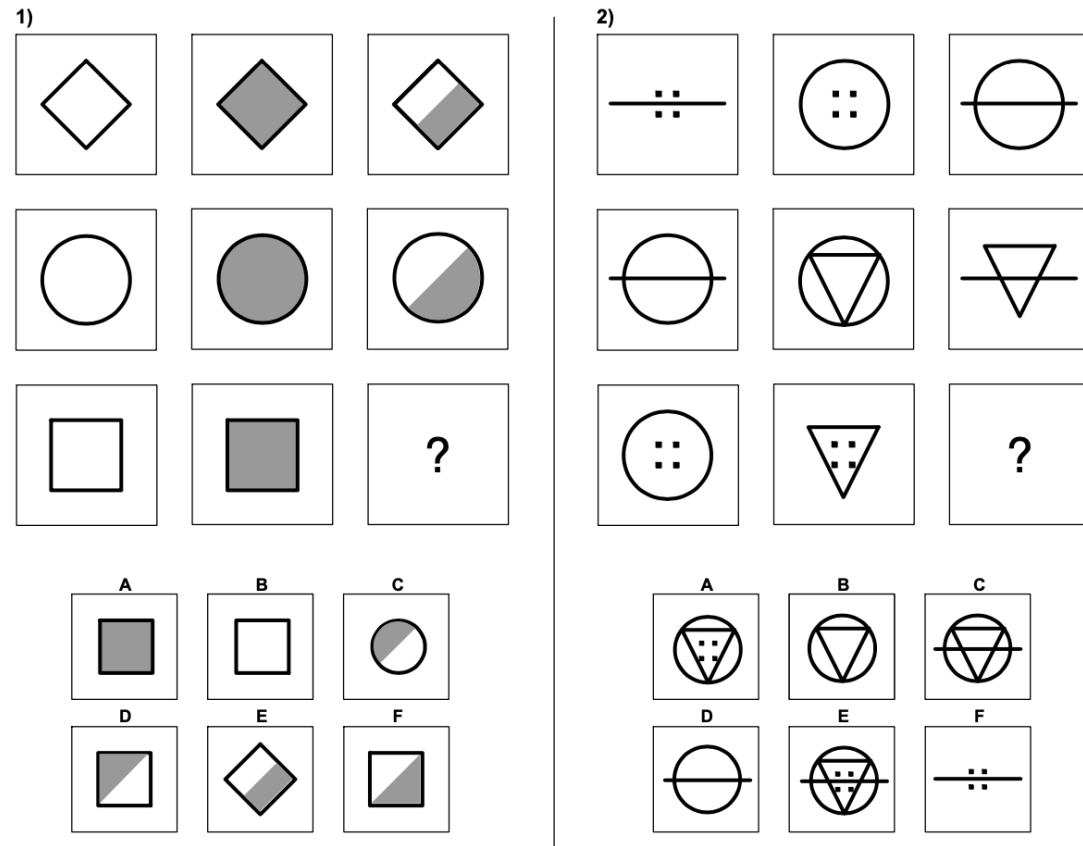
- Ceiling effects
- Insufficient detail for cognitive domains

Cognitive Domains

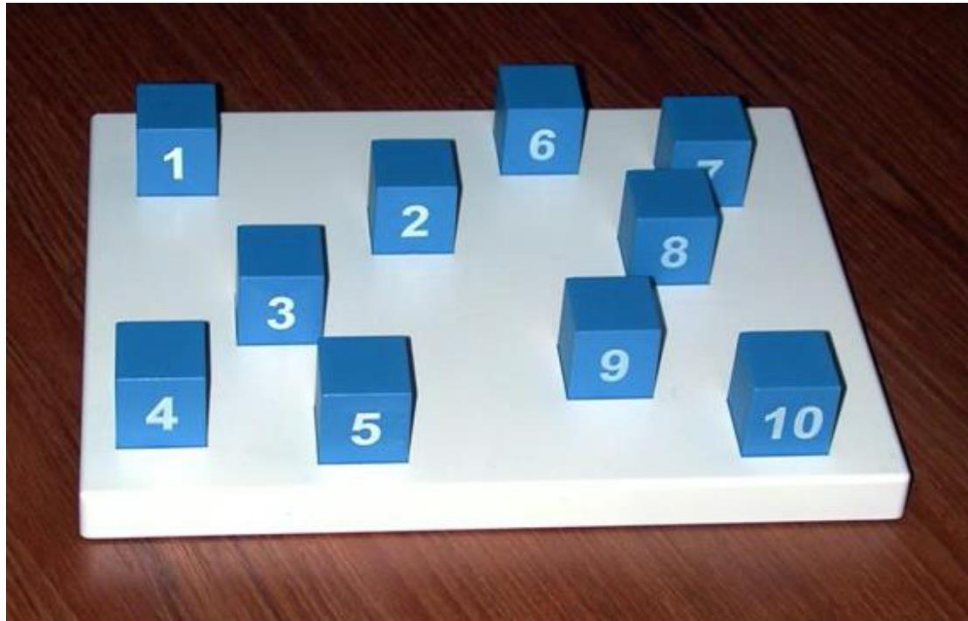
Vocabulary

ZEALOT
BANAL
CATACOMB
GAOLED
FACADE
CELLIST
TOPIARY
QUADRUPED
ABSTEMIOUS
GAUCHE
AVER
DETENTE
IDYLL
RADIX
AEON
PUERPERAL
BEATIFY
LEVIATHAN

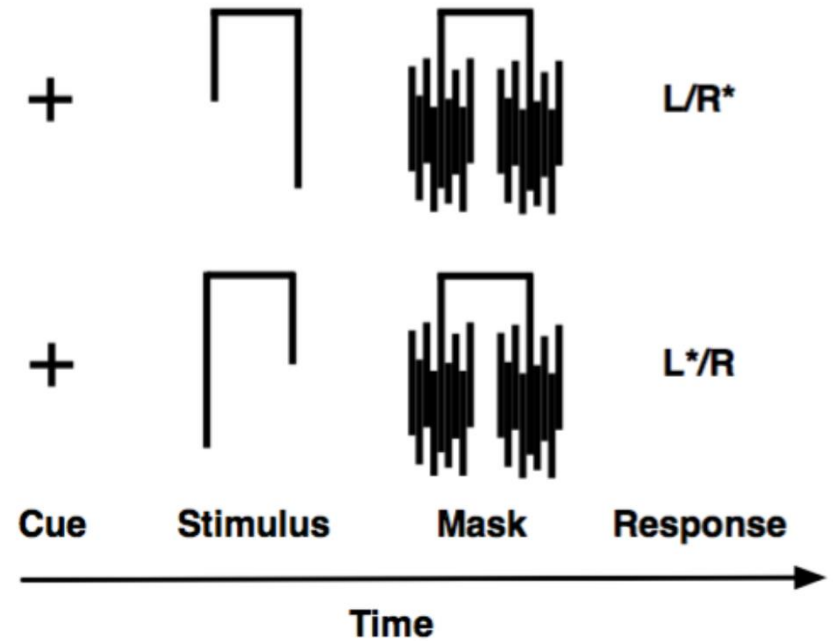
Reasoning



Memory



Processing Speed



Cognitive domains

Vocabulary

ZEALOT
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PUERPERAL
BEATIFY
LEVIATHAN

Reasoning

1)

A

B

C

D

E

F

2)

A

B

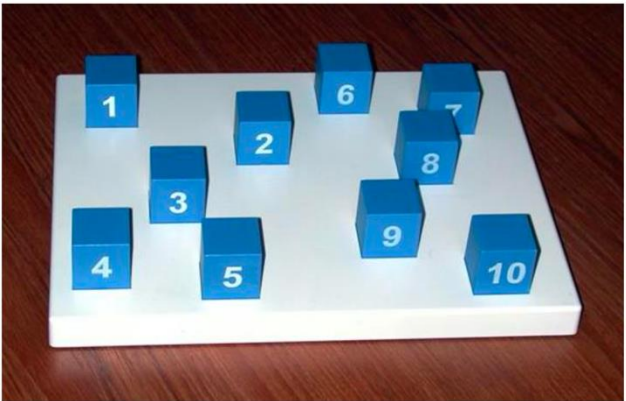
C

D

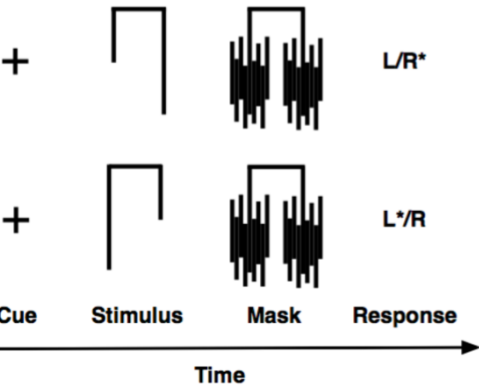
E

F

Memory



Processing Speed

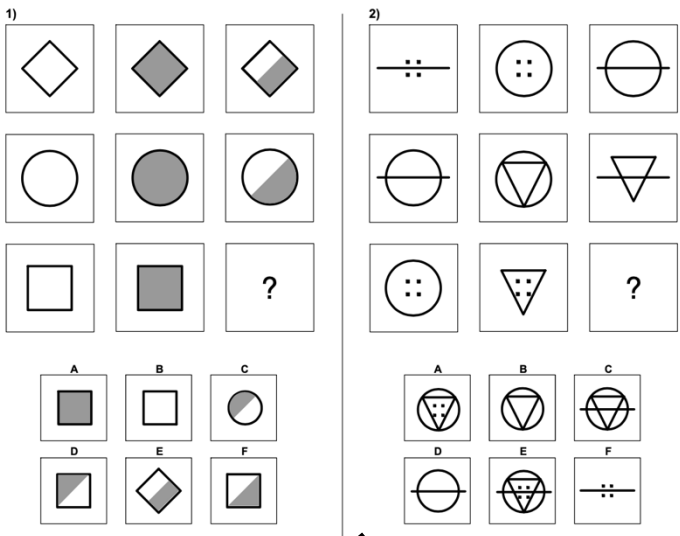


Cognitive domains

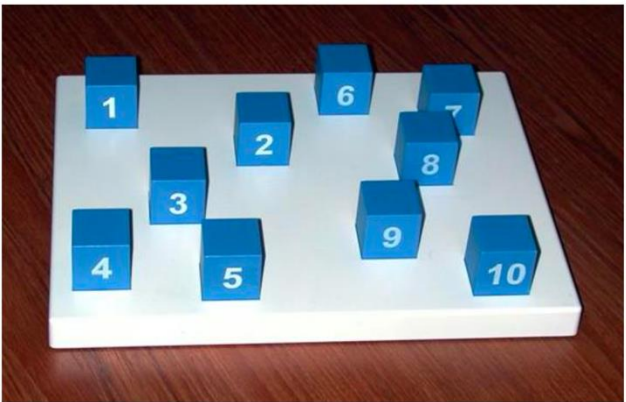
Vocabulary

ZEALOT
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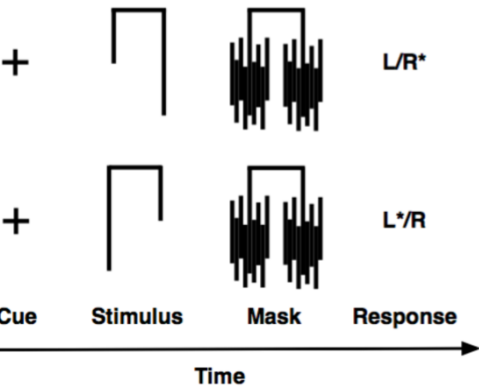
Reasoning



Memory



Processing Speed



?

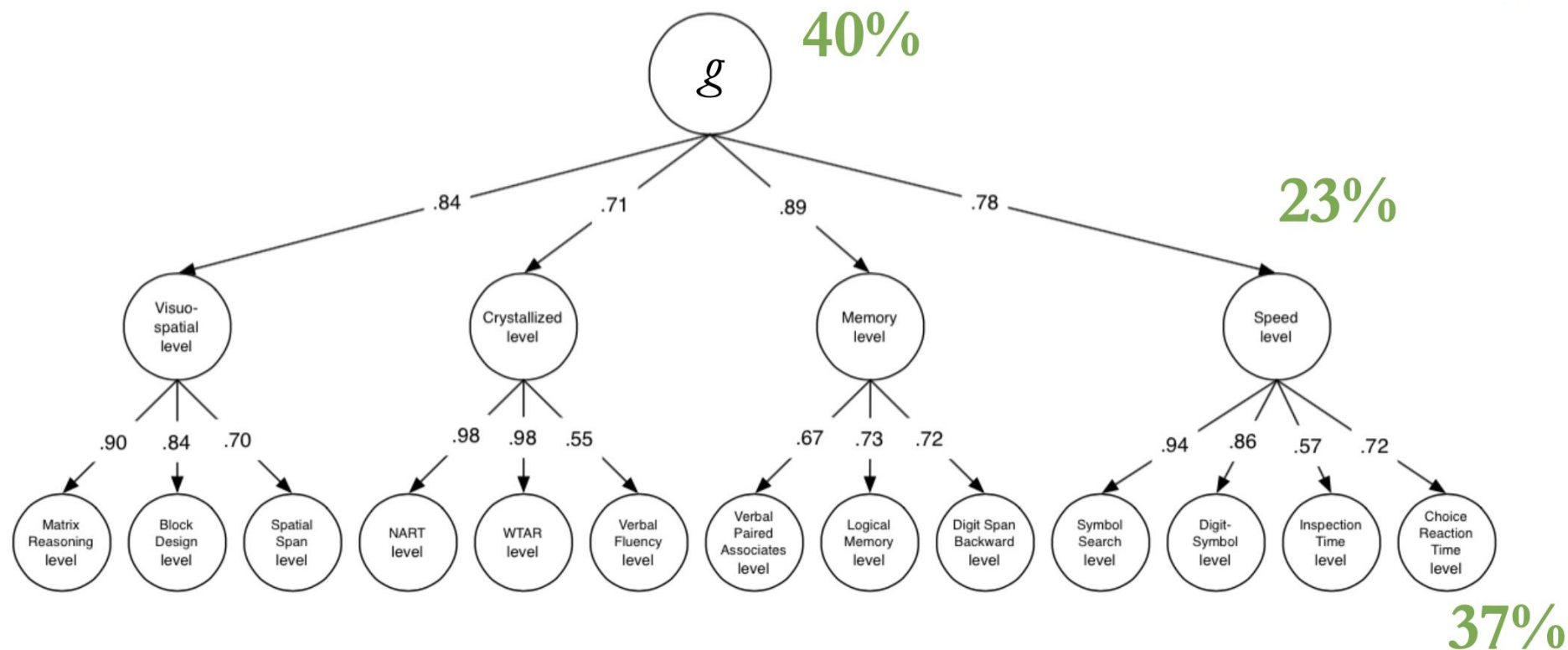
***g* : general cognitive function**
[much of our cognitive differences are general]



Fluid vs crystallized intelligence?

Susceptibility to ageing?

Nothing but a *g* thing?

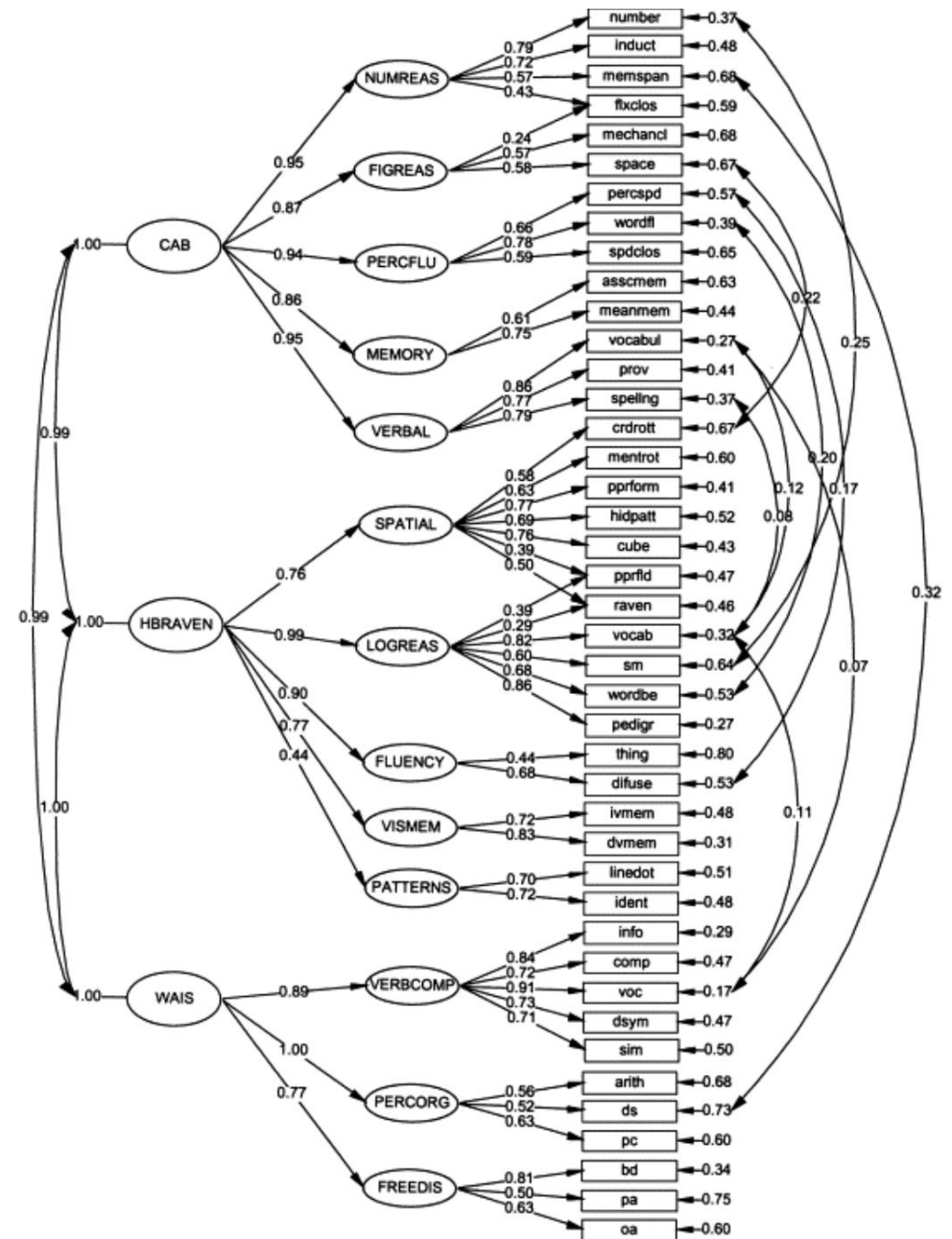


Robust and well-replicated

Johnson et al – Just one g

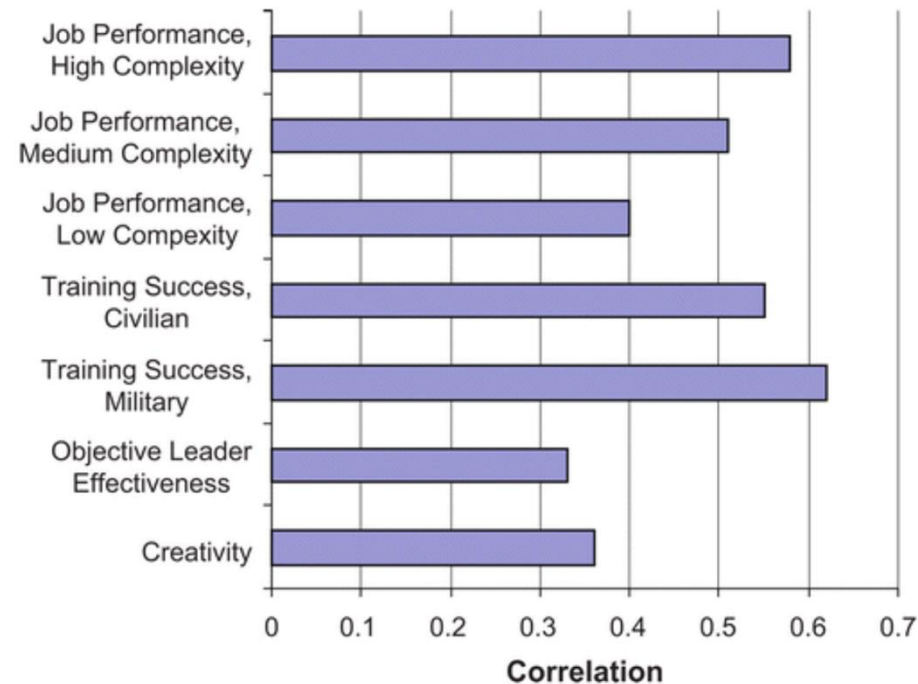
- Administered 3 different test batteries.
- Variety of cognitive abilities.
- $N = 436$ (188 M, 248 F).
- Age range 18-79yrs ($M = 42.7$ yrs).

g correlations all ≥ 0.99



Where does a high g get you?

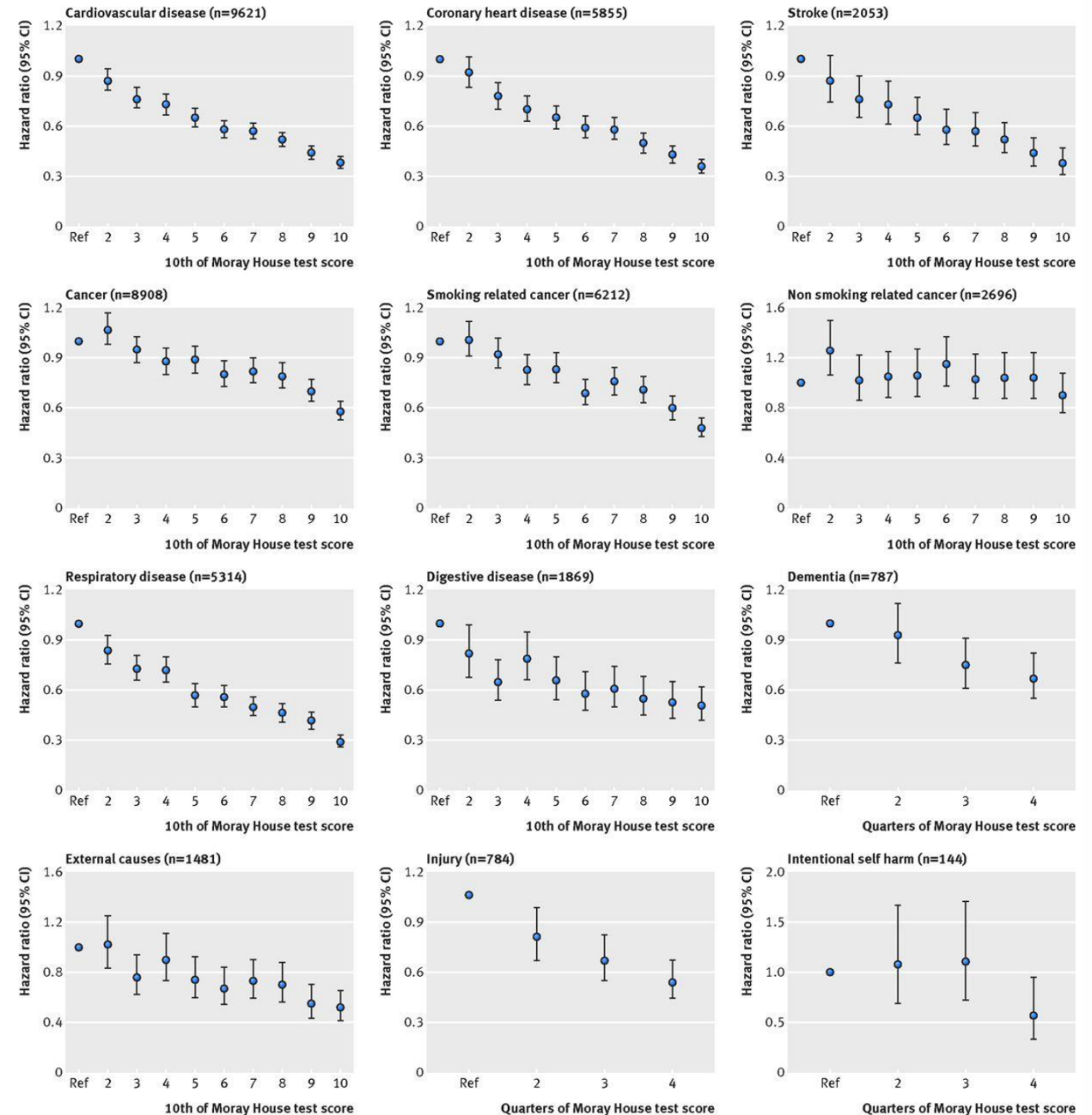
Job performance



Mortality

- Age 11 intelligence (SMS '47)
- Lifecourse mortality records up to 79yrs.
- N = 65,765

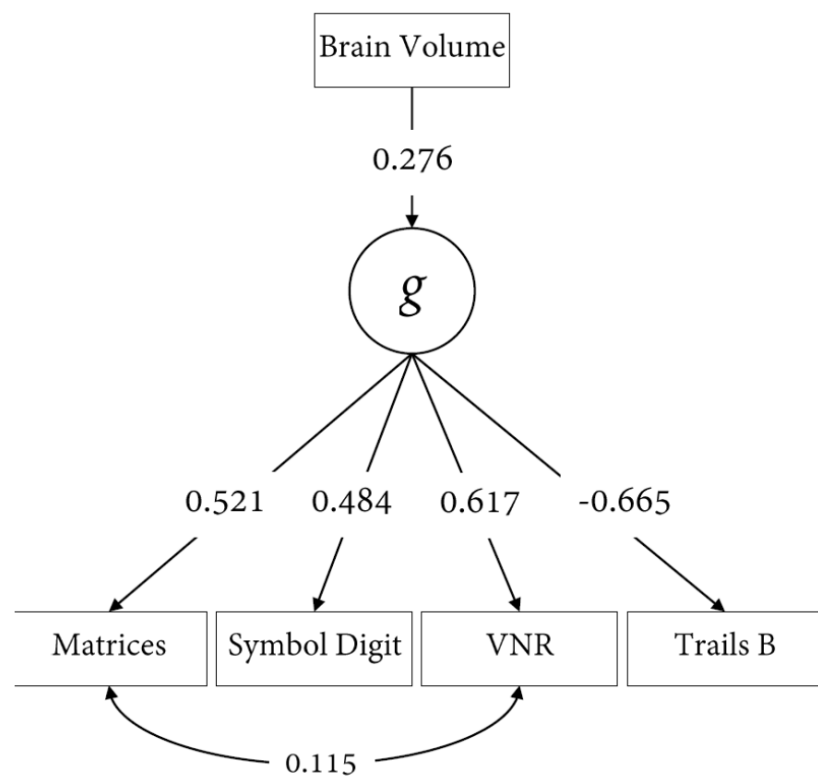
Results Childhood intelligence was inversely associated with all major causes of death. The age and sex adjusted hazard ratios (and 95% confidence intervals) per 1 SD (about 15 points) advantage in intelligence test score were strongest for respiratory disease (0.72, 0.70 to 0.74), coronary heart disease (0.75, 0.73 to 0.77), and stroke (0.76, 0.73 to 0.79). Other notable associations (all $P < 0.001$) were observed for deaths from injury (0.81, 0.75 to 0.86), smoking related cancers (0.82, 0.80 to 0.84), digestive disease (0.82, 0.79 to 0.86), and dementia (0.84, 0.78 to 0.90). Weak associations were apparent for suicide (0.87, 0.74 to 1.02) and deaths from cancer not related to smoking (0.96, 0.93 to 1.00), and their confidence intervals included unity. There was a



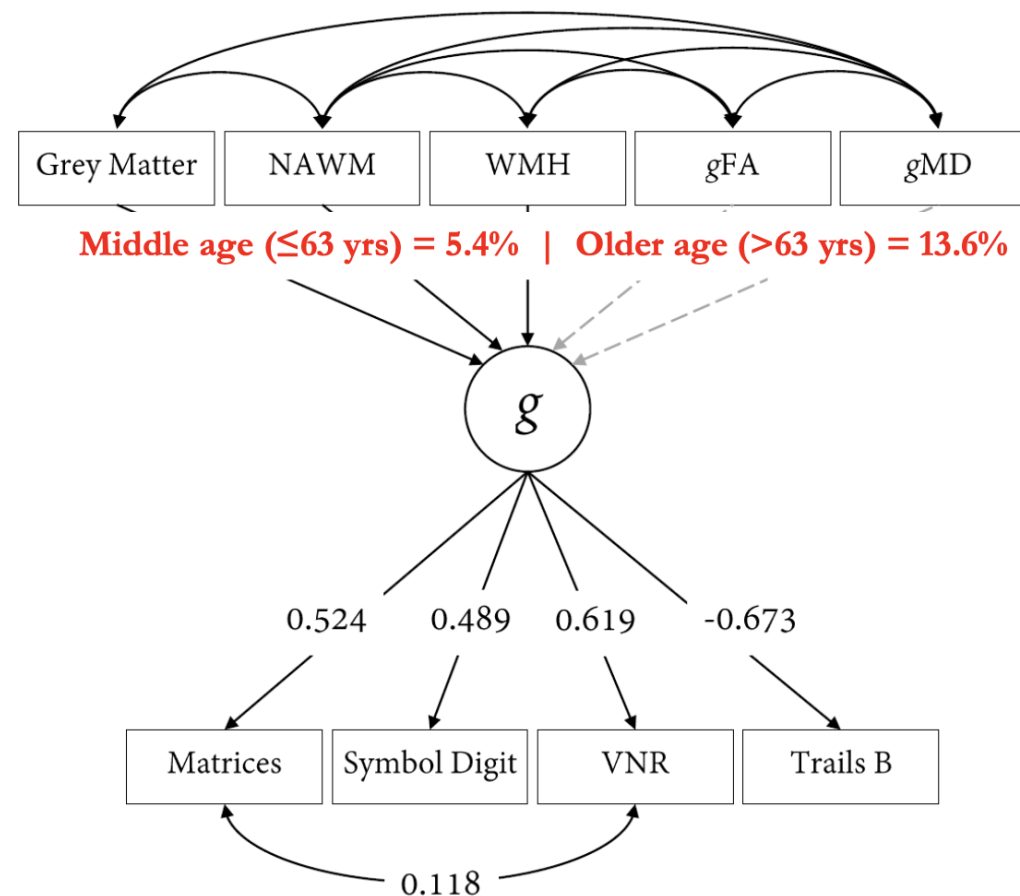
Brain biomarkers of cognitive function, and cognitive ageing?

- Global brain correlates of general cognitive ability.

a

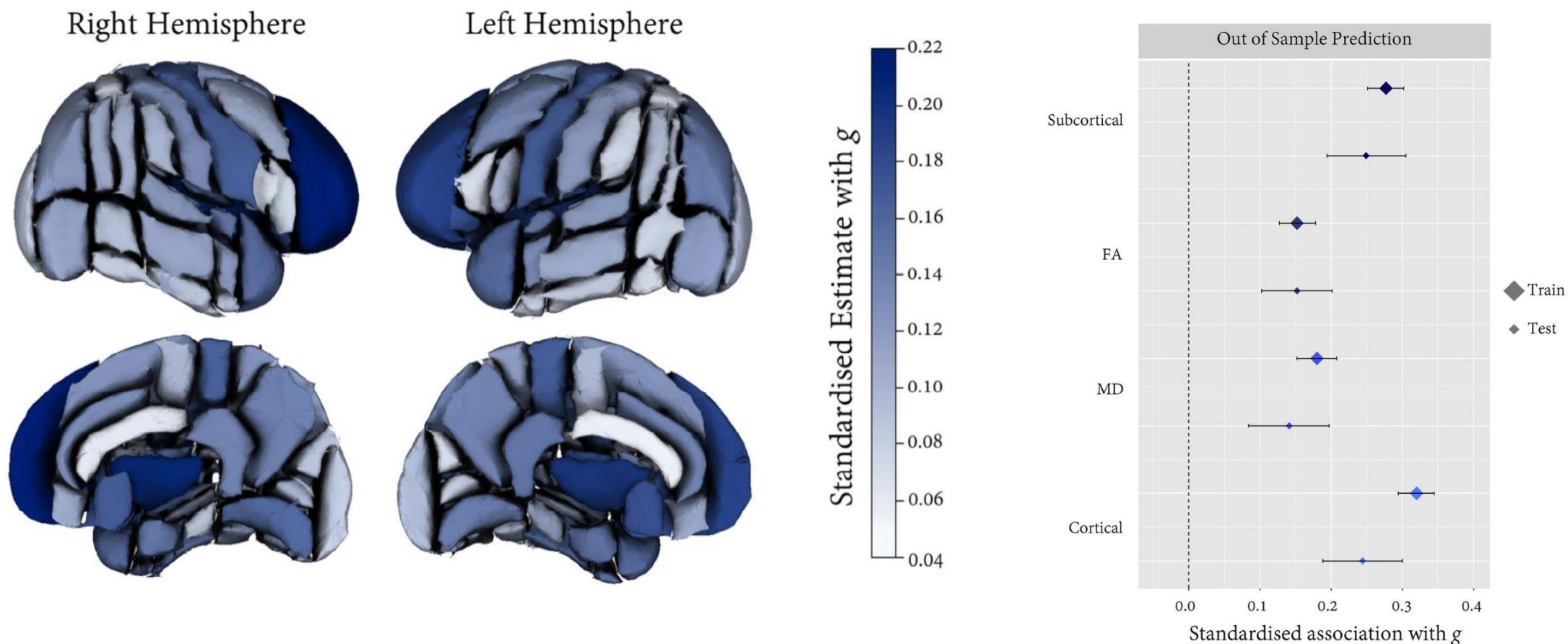


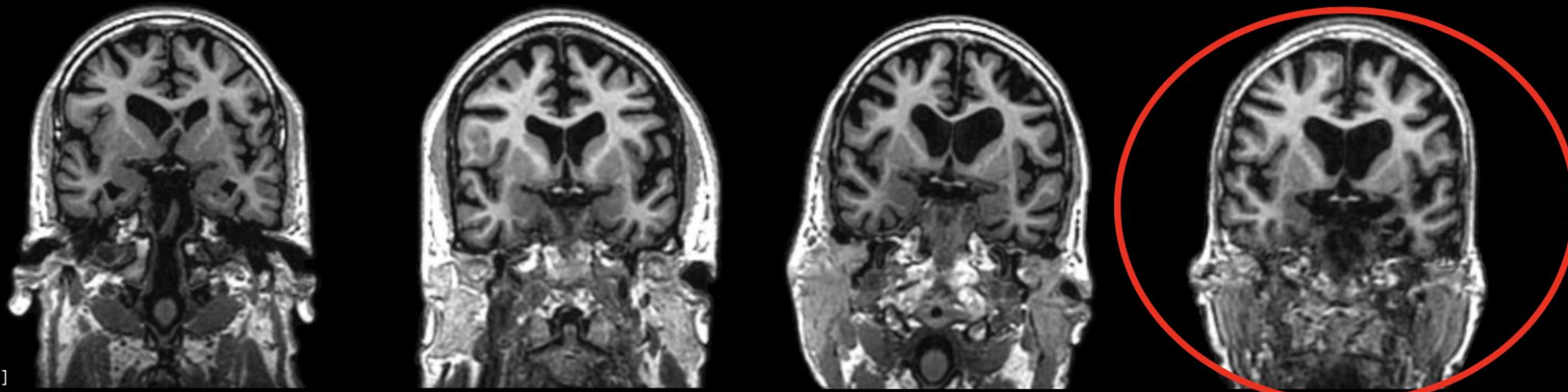
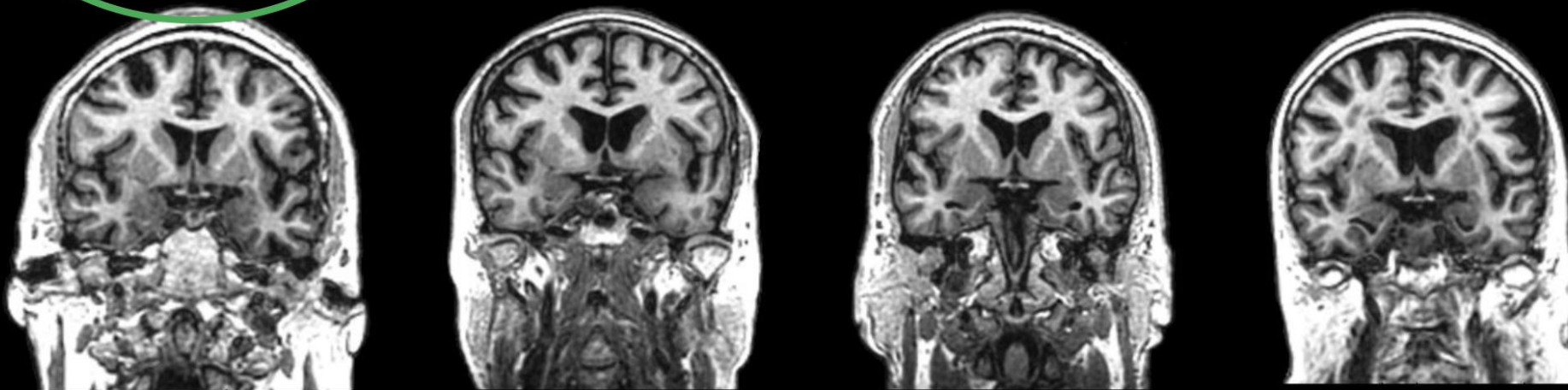
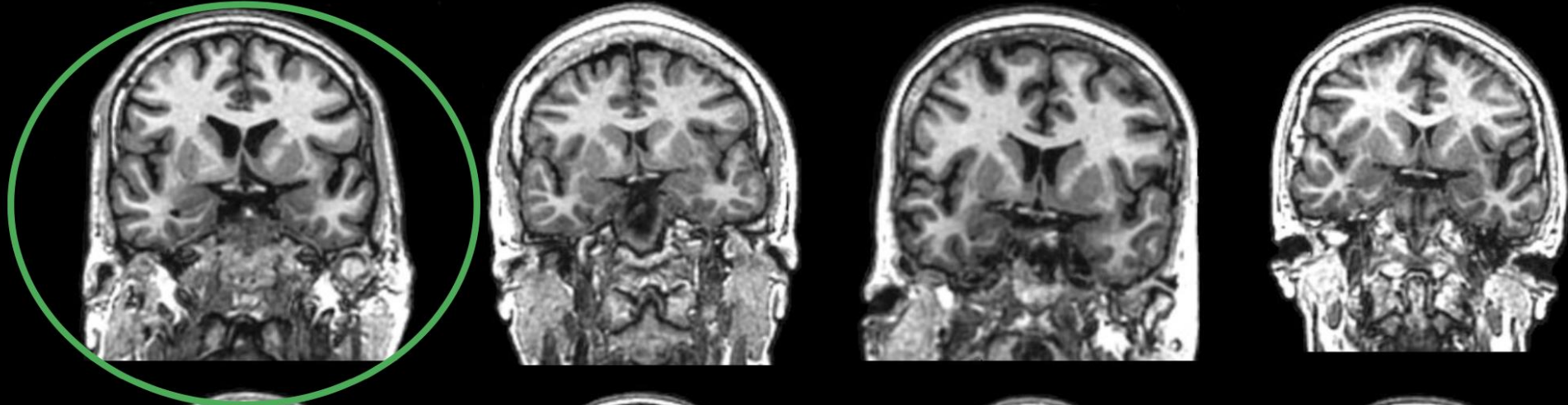
b

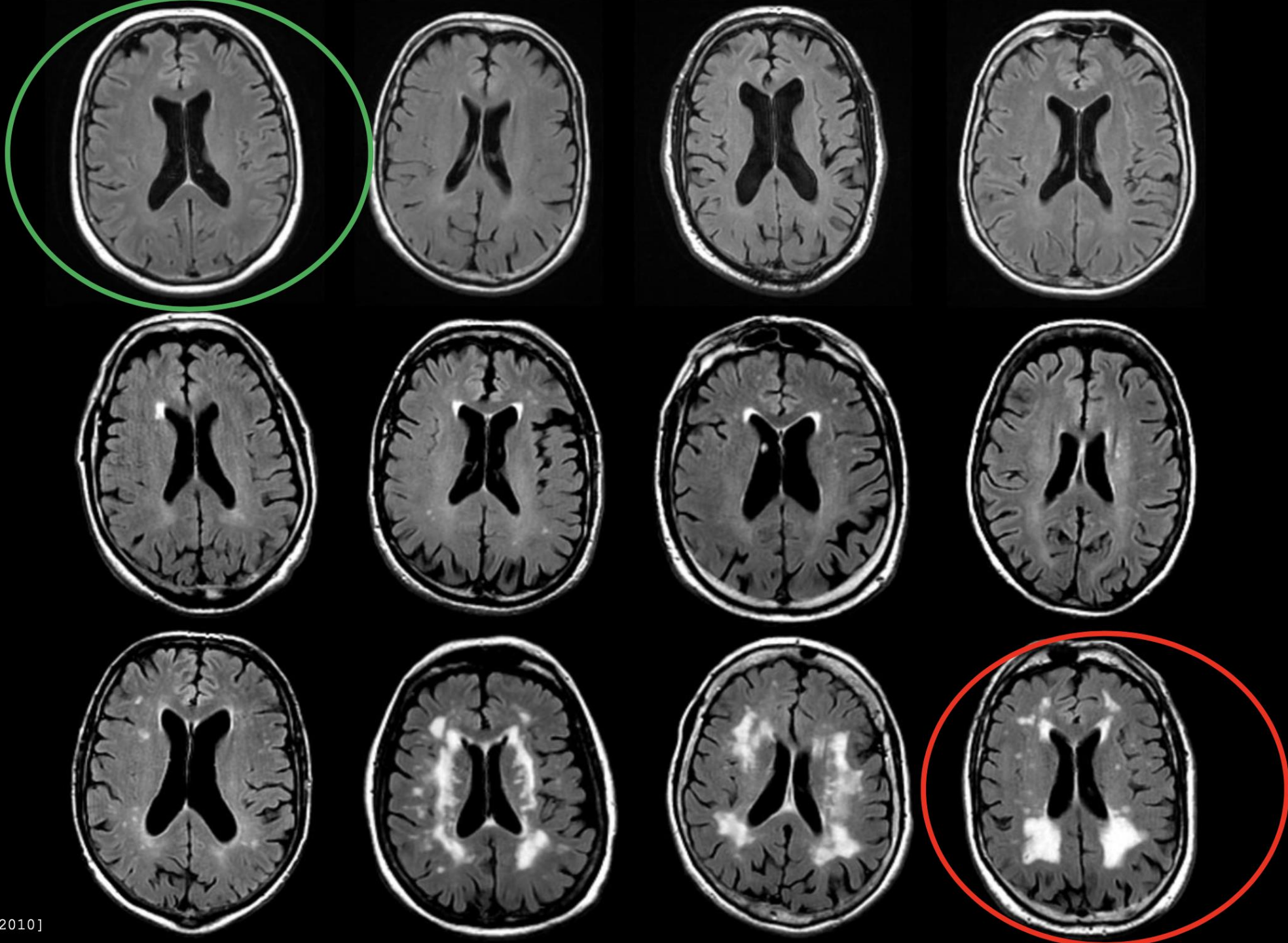


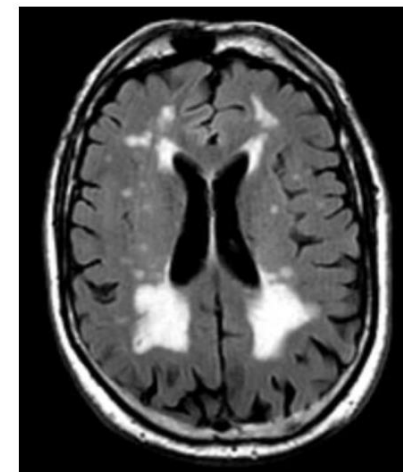
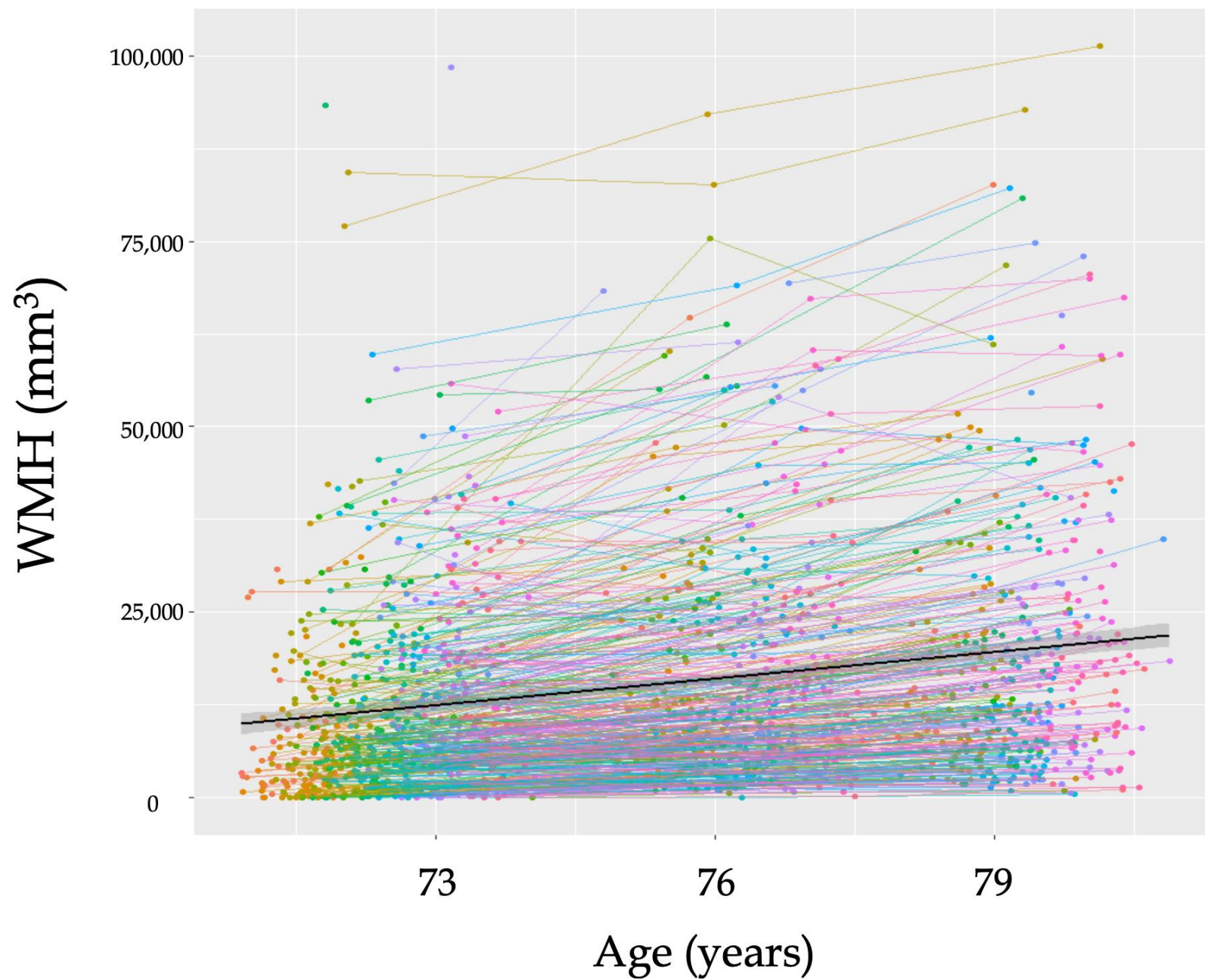
Brain biomarkers of cognitive function, and cognitive ageing?

- Regional grey matter correlates of general intelligence.









Stages of memory

Sensory

Ichonic

Short-term

Long-term

Stages of memory

Sensory

Iconic = VISUAL

(visual) sensory memory --
can be experimentally
studied by using the partial
report technique
(Coltheart, 1980)

Short-term

Long-term

Stages of memory

Sensory

Ichonic

Short-term

Long-term

What are the problems associated with reliability of memory?

Aka False Memories

Memory

False Memories

- **Eyewitness testimony** is unreliable due to the susceptibility of memory to **suggestion, leading questions, and emotional bias**

Memory

False Memories = Reconstructive

- **Misinformation effect:** False memories arise when **new information** contaminates retrieval.
- **Source monitoring errors** (Johnson, 1993): Difficulty distinguishing real from imagined events.
- **Hippocampus activation** in both true & false memories (Schacter, 2001).